

Macprudential Policy with Firm Heterogeneity

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Abstract

I study optimal macroprudential policy when its effects on investment and productivity are taken into account. To do so, I introduce a tractable way of modeling misallocation that generates a link between investment and productivity and can be easily taken to the data. Because macroprudential policies affect investment, they lead to productivity losses. I show that, when the policymaker is constrained in their available instruments, this generates a policy trade-off between financial stability and productivity growth. I derive a sufficient statistics formula for the second-best policy, including its productivity costs. I leverage the tractability of my model to get a range of estimates for the latter using rich firm-level microdata for several European countries. The trade-off is quantitatively relevant: For baseline crisis probabilities, productivity losses switch optimal policy from a capital control to a foreign borrowing subsidy.

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1 Introduction

Macroprudential policies have become a key policy tool in recent years, as shown by the introduction of Basel III and their recognition as a valid policy instrument by the International Monetary Fund (2022).¹ They are supported by an extensive literature (Bianchi and Mendoza, 2020), which focuses on their role in ensuring financial stability. For small open economies, in which foreign borrowing is a significant source of credit, these policies usually take the form of capital controls.

At the same time, there has been increasing attention to the role of international capital flows in alleviating (Forbes, 2005; Larrain and Stumpner, 2017; Varela, 2017; Bau and Matray, 2023) or exacerbating misallocation (Gopinath et al., 2017; Benigno and Fornaro, 2014). This role introduces an additional consideration to the study of macroprudential policy.

In this paper, I examine how optimal macroprudential policy changes when these two forces are present. To do so, I build on the seminal open economy model with borrowing constraints developed by Bianchi (2011), to which I introduce investment and capital misallocation across firms. I introduce a tractable way of modeling misallocation that generates a link between investment and productivity and has a clear mapping to the data. I leverage this feature to estimate the elasticity of productivity to investment for a sample of European countries using firm-level microdata.

Using this model, I first study the determinants of optimal macroprudential policy. On the one hand, pecuniary externalities mean that households do not fully internalize the costs of their consumption, which leads them to overborrow. Capital controls can correct this, as is standard in the literature. On the other hand, by increasing borrowing costs, they reduce investment and productivity. When the government can separately tax or subsidize borrowing and investment, these forces generate no trade-off. I show that the planner implements capital controls to curb borrowing and subsidizes investment to ensure that households invest the optimal amount.

I then show that a trade-off exists when the government can control the total amount of borrowing using capital controls but cannot alter its composition between investment and consumption. In this second-best scenario, I show that the policymaker faces a trade-off between restricting borrowing to reduce consumption and incentivizing borrowing to increase investment.

I derive a sufficient statistic formula for this capital control, which depends on easily measurable elements. I show how each of these affect optimal policy and how to measure

¹See Cerutti, Claessens, and Laeven (2017) for a survey of their prevalence across countries.

them using available data. The second-best capital control depends on the probability of a crisis and the magnitude of the productivity losses that result from capital controls. I use the range of productivity losses I estimate in the data and find that, when the probability of a crisis is low, the second-best capital control, implemented as a tax on debt, turns into a borrowing subsidy as high as -1.25%. As a crisis becomes more likely, the tax is as high as 3.4%.

My starting point is the canonical model of macroprudential policy developed by Bianchi (2011). Households derive utility from consuming tradable and non-tradable goods. The former can be traded with the rest of the world, along with non-contingent financial claims. Households own an endowment of non-tradable goods and supply labor and capital to tradable goods producers that produce individual varieties which are aggregated by competitive firms.

Households are subject to a borrowing constraint that depends on their income measured in units of tradable goods. When the constraint binds, households are forced to deleverage by reducing their consumption of tradable goods. This also reduces demand for non-tradable goods, pushing their price down and reducing the market value of the household's income. This further tightens the borrowing constraint, which gives way to another round of deleveraging in a vicious cycle. These dynamics capture the stylized facts of sudden stops, as financial crises with large current account reversals are known.

The pecuniary externality arises because households do not internalize the fact that increasing the consumption of tradable goods increases the value of non-tradable goods, and therefore their ability to borrow when the borrowing constraint binds. By increasing consumption during a crisis, macroprudential policies that reduce borrowing during good times can be welfare improving. An example of these policies are capital controls, which increase the cost of foreign borrowing.

Because the model studied by Bianchi (2011) and much of the literature does not consider production economies, they do not capture the effects of macroprudential policy on capital accumulation and productivity. To study the qualitative and quantitative implications of this cost, I add the production of tradable goods using labor and capital. To further examine the productivity implications, I consider a setting in which firms produce individual varieties that are aggregated with a CES technology. As shown by Hsieh and Klenow (2009), this allows a clear mapping from the distribution of distortions at firm level to total factor productivity.

I model misallocation by introducing firm-level frictions in the rental market for capital. Domestic banks intermediate between households and firms. The amount of capital firms can rent from a bank is subject to distortions, which leads to dispersion in the marginal return to capital across firms. I assume that these distortions do not grow proportionally with capital,

which implies that an increase in the aggregate amount of capital reduces misallocation in the economy. This is in line with the literature on the link between financial development and growth (Rajan and Zingales, 1998; Levine, 2005) and with the experience of countries that lifted capital controls (Larrain and Stumpner, 2017; Varela, 2017) or restrictions on investment (Bau and Matray, 2023).

I then derive a closed-form expression for the link between total factor productivity and aggregate capital, which resembles a production externality. In this setting, capital controls have two adverse effects on output. First, the increase in borrowing cost raises the required rate of return on capital, which leads to a decrease in investment. Second, reduced investment, through increased misallocation, negatively affects productivity, which means that the same amount of capital produces fewer goods.

As a benchmark, I first study the problem of a policymaker who is not constrained with respect to available instruments. Because they can tax borrowing and investment separately, they face no trade-off. Capital controls address the pecuniary externality and reduce consumption when the economy is not in a crisis, while a subsidy to investment addresses the productivity externality.

I then turn to a more realistic setting, in which the policymaker can only control the total level of debt but not its composition between investment and consumption. In this scenario, I find that the planner faces a trade-off between the two objectives: financial stability and productivity growth. The former requires a capital control to reduce consumption during good times, while the latter needs a subsidy on foreign borrowing to encourage investment and productivity growth.

To gain more intuition, I show how the second-best capital control depends on a key sufficient statistic composed of four measurable objects: the probability of a sudden stop, the productivity externality, the marginal propensity to invest in tradable sectors and the elasticity of substitution between tradable and non-tradable goods.

The probability of a sudden stop determines the importance of the macroprudential component. When a crisis is more likely, capital controls should increase to deter borrowing. I consider a range of values for this probability, from the unconditional probability for developed economies (1.7%) to the highest probability in the “red-zones” framework of Greenwood et al. (2022) (37%–45%). For the former, the policymaker subsidizes borrowing for plausible parameters but imposes strong capital controls for the latter, with interest rates increasing by almost 3.5 percentage points.

The productivity externality is one of the main drivers of the policy trade-off. In one of the key contributions of the paper, I leverage its closed-form expression and estimate a range of plausible values using microdata. Using firm-level balance sheet data from Orbis-

Amadeus, which has excellent coverage of private European firms, I measure the dispersion in the marginal returns to capital across firms, which directly informs my estimates. I estimate an elasticity of total factor productivity to investment that ranges between 0.05 and 0.13 for a sample of European economies. In the baseline scenario, in which a crisis is not likely, this implies that economies at the top of the range should have a subsidy three times as large as economies at the lower end of the range.

The marginal propensity to invest in tradable production is a key object when the government cannot control the allocation of credit. For low values, most of the increase in credit will finance consumption, which is what the policymaker wants to avoid. Lastly, the elasticity of substitution between tradable and non-tradable goods determine the strength of the pecuniary externality. A reduction in debt during good times increases available resources during a sudden stop. The marginal propensity to consume indicates how this translates to tradable consumption.

I find that the investment and productivity costs have significant effects on the second-best capital control. Evaluated at the unconditional probability of a sudden stop, my formula predicts a 0.5% tax on foreign borrowing when these effects are not taken into account. Once they are considered, the tax turns into a subsidy of almost 0.4% for the median country in the sample, with almost two-thirds of this reduction driven by effects on productivity. Moreover, the latter are a significant source of heterogeneity, generating a maximum difference of 0.35% in capital controls between countries in the sample.

Related Literature. This paper connects two strands of the literature. The first studies optimal macroprudential policy in a context of pecuniary externalities and financial frictions that render the competitive equilibrium inefficient. The second strand studies how frictions in factor markets, through misallocation of resources across firms, lead to reduced aggregate productivity.

Most of the macro-finance literature on macroprudential policy builds on the insights of Bernanke and Gertler (1989) and Kiyotaki and Moore (1997), who demonstrate how financial frictions can amplify the effect of shocks, formalizing an old intuition (Fisher, 1933). The combination of incomplete markets and the role prices play in them lead to the emergence of pecuniary externalities, as studied by Clayton and Schaab (2022), Dávila and Korinek (2017), Gromb and Vayanos (2002) and Lorenzoni (2008). In many cases, these externalities lead to inefficiencies, which motivate the study of policy interventions.² This normative literature, starting with Jeanne and Korinek (2010) and Bianchi (2011), has mostly focused

²Schmitt-Grohé and Uribe (2016) study macroprudential policy in the context of wage rigidities and currency pegs.

on small open economies.³

The main contribution of this paper is to build upon the seminal setup of Bianchi (2011), which has been extensively used (Benigno, Chen, et al., 2013; Arce, Bengui, and Bianchi, 2019; Flemming, L’Huillier, and Piguillem, 2019; Seoane and Yurdagul, 2019; Schmitt-Grohé and Uribe, 2020; Bengui and Bianchi, 2022) to study how effects on productivity affect optimal macroprudential policy.

I make two contributions to this literature. The first is by considering both capital accumulation and endogenous productivity, which increases the costs of macroprudential policy in a second-best setting. In this regard, the model I study resembles the one in Bianchi and Mendoza (2020), which also features capital accumulation but has exogenous productivity. By analytically characterizing second-best policy in this setting, I show that considering these additional features affects optimal macroprudential policy.

This also demonstrates the paper’s second contribution to this literature, which is to characterize both the constrained-efficient and second-best policy and leverage this characterization to provide a quantification of the trade-off as a function of measurable sufficient statistics. This approach is similar to that of Dávila and Korinek (2017), who also characterize macroprudential policy using sufficient statistics. Compared with their work, I find a direct mapping to the data which allows me to quantify the trade-off.

I also draw on the findings of a recent literature that examines the effects of capital controls on productivity. Larrain and Stumpner (2017) and Varela (2017) study episodes in which restrictions on capital mobility were lifted in Eastern European countries, while Bau and Matray (2023) study the easing of restrictions on foreign equity investments in India. Alfaro, Chari, and Kanczuk (2017) use Brazilian stock market data to analyze the effect of capital controls on firms. All find that easing controls reduces the cost of capital more for constrained firms, which results in increased aggregate productivity through improved allocation of capital. These results are also in line with an earlier literature surveyed by Forbes (2005).⁴

In studying the credit boom in Southern Europe during the introduction of the euro, Gopinath et al. (2017) find a decrease in productivity in the manufacturing sector through increased misallocation.⁵ This would suggest that capital inflows can have negative effects on productivity, in line with a theoretical literature on credit booms (Asriyan et al., 2024;

³Noteworthy exceptions are Farhi and Werning (2016) and Elenev, Landvoigt, and Van Nieuwerburgh (2021), who study the effects of macroprudential policies in a quantitative model of a closed economy.

⁴This is also consistent with Madeira (2024), who examines the effects of macroprudential policies on industry growth and finds that financial frictions play a key role in understanding the real effects of macroprudential policy.

⁵See also Calligaris et al. (2018) and Dias, Robalo Marques, and Richmond (2016)

Benigno, Chen, et al., 2016; Reis, 2013). Studying a large panel of countries Monacelli, Sala, and Siena (2023) find that increases in interest rates lead to reductions in aggregate productivity in emerging economies, while the opposite is true for emerging economies. Although in the quantitative exercise I consider a positive effect of capital inflows on productivity, this paper still provides a framework to consider these effects when studying optimal macroprudential policy.

These papers are part of a broader literature⁶ on how the misallocation of factors across firms reduces aggregate productivity. I follow the indirect measuring framework introduced by Hsieh and Klenow (2009), which allows me to link model and data in a tractable manner. This approach was also used by Midrigan and Xu (2014), David and Venkateswaran (2019) and David, Venkateswaran, et al. (2021) to decompose the sources of misallocation, and find that technological frictions can explain only a small share of total misallocation between firms. I contribute to this literature by embedding this setup in an optimal policy analysis that takes these empirical results into account.

In that regard, the paper is related to a growing literature that examines the effects on productivity through misallocation of different policies. Baqaee, Farhi, and Sangani (2024) and González et al. (2023) study how misallocation affects the transmission of monetary policy, and Kurtzman and Zeke (2020) show that quantitative easing can increase misallocation. David and Zeke (2024) study how the monetary policy regime shapes the dynamics of TFP and how optimal monetary policy is affected by this mechanism. This paper is related to theirs with respect to letting the data guide the quantification of the effects of policy on productivity.

Lastly, Andreasen et al. (2023) use a quantitative model to study the effects of the Chilean “encaje”, a form of capital control, on productivity. While the specifics of their model differ, they share the conclusion that capital controls decrease TFP. Calibrated to the Chilean example, they find welfare losses of up to 0.61% in consumption-equivalent units. Because they do not consider aggregate risk, they cannot speak to the trade-off inherent to macroprudential policy that I explore in this paper.

Layout The remainder of the paper is organized as follows. Section 2 introduces the model that will guide my analysis while Section 3 defines and characterizes the equilibrium and discusses the inefficiencies that motivate studying optimal policy. Section 4 sets up and characterizes optimal policy, and Section 5 studies its determinants. In Section 6, I leverage the results from Section 4 to present quantitative results. Section 7 concludes.

⁶See, e.g., Restuccia and Rogerson (2013) and Hopenhayn (2014).

2 Model

To study the policy trade-off, I build on the canonical model presented in Bianchi (2011) to which I add production in the tradable sector along with financial frictions that differ across firms. Aggregate risk is given by the realization of tradable productivity and a stochastic tightening of the borrowing constraint.

2.1 Environment

I study a small open economy composed of households, firms that produce final and tradable goods and domestic banks that intermediate between households and firms. Households trade bonds and goods with the rest of the world, while firms can export tradable goods.

Households. This block follows closely the setup in Bianchi (2011) with the addition of capital goods. Households have preferences over a final consumption good, c_t , described by

$$\mathbb{E}_0 \left[\sum_t \beta^t \frac{c_t^{1-\sigma} - 1}{1-\sigma} \right] \quad (1)$$

Households are able to borrow, or save, through a one period, risk-free bond d_{t+1} denominated in units of tradable goods traded with the rest of the world at an exogenous rate r . They also accumulate physical capital k_t , which is rented to tradable firms. Lastly, households receive a fixed endowment of non-tradable goods, y^N , and are the owners of the tradable firms.

I assume that the tradable good is the numéraire, so that p_t is the relative price of final goods, and p_t^N that of non-tradable goods. The budget constraint of the households, omitting references to the state of the economy, is then given by

$$p_t c_t + p_t (k_{t+1} - (1 - \delta)k_t) = w_t + (R_t + \tau_t^k)k_t + p_t^N y^N - (1 + r)(1 + \tau_t^d)d_t + d_{t+1} + \pi_t + T_t \quad (2)$$

where π_t are profits from firms, w_t is the wage rate and R_t is the rental rate of capital. There are two policy instruments: a tax on debt, or capital control, τ_t^d and an investment subsidy τ_t^k . I assume the government sets a lump-sum T_t to balance its budget.

As is common in the macroprudential literature, households also face a borrowing constraint that depends on the market value of their income.

$$d_{t+1} \leq \kappa (y_t^T + p_t^N y^N) - \bar{d}_t, \quad (3)$$

where $\kappa < 1$ is a constant and $\bar{d}_t \geq 0$ is a stochastic variable that captures exogenous fluctuations to the borrowing capacity of households⁷.

The Euler equation for debt d_{t+1} , is given by

$$\frac{c_t^{-\sigma}}{p_t} = \frac{1}{1 - \mu_t} \beta(1 + r)(1 + \tau_t^d) \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right], \quad (4)$$

where μ_t is the Lagrange multiplier on the borrowing constraint (3)⁸. As usual, the households seek to smooth consumption between periods. The capital control τ_t^d shows up as a wedge in this condition, altering the interest rate faced by the households. When the constraint binds, $\mu_t > 0$ acts like a wedge in this Euler equation. To see this, re-arrange (4) as

$$\mu_t = 1 - \beta(1 + r)(1 + \tau_t^d) \mathbb{E} \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \right] \quad (5)$$

Expressed in this way, μ_t measures the distance from the desired consumption smoothing when the borrowing constraint binds. Within the model, I consider periods where the borrowing constraint binds to be a crisis or sudden stop. In this way, μ_t serves both as an indicator of whether a crisis is happening, $\mu_t > 0$, and its severity, measured as the drop in consumption relative to a non-crisis state.

Lastly, the Euler equation determining investment is given by

$$c_t^{-\sigma} = \beta \mathbb{E} \left[c_{t+1}^{-\sigma} \left(\frac{R_{t+1}}{p_{t+1}} (1 + \kappa \mu_{t+1}) + \frac{\tau_{t+1}^k}{p_{t+1}} + 1 - \delta \right) \right] \quad (6)$$

Ignoring μ_t , this is the standard condition for optimal investment. The presence of crises has two effects on the capital accumulation decision. First, when the constraint is presently binding, $\mu_t > 0$, this raises the effective interest faced by the household, increasing the required rate of return on capital, lowering investment. When the constraint is expected to bind in the next period, $\mu_{t+1} > 0$ the effect is the opposite. Because households understand that more income will increase their borrowing ability, they derive an additional benefit from investing an extra unit of capital, given by $\kappa \mu_{t+1}$.

Conditions (10), (4), (6), along with the budget constraint (2), the borrowing constraint (3), and transversality and complementary slackness conditions characterize the solution to the household's problem.

⁷Including this additional shock helps matching the dynamics of sudden stops in the quantitative exercise but does not alter the main results of the paper.

⁸Normalized by the Lagrange multiplier on the budget constraint

Final Goods Producers A representative firm combines tradable and non-tradable inputs to produce final goods x_t to sell to households according to:

$$x_t = \left[\omega x_{T,t}^{\frac{\xi-1}{\xi}} + (1-\omega) x_{N,t}^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}} \quad (7)$$

where $x_{T,t}$ and $x_{N,t}$ denote tradable and non-tradable inputs, respectively.

The problem of this firm is given by

$$\max_{x_t^T, x_t^N} p_t x_t - x_t^T - p_t^N x_t^N \quad (8)$$

The first order condition with respect to x_t^T pins down the relative price of final goods

$$p_t = \frac{1}{\omega} \left(\frac{x_t}{x_t^T} \right)^{\frac{1}{\xi}} \quad (9)$$

Combining this condition with the first order condition with respect to x_t^N yields the intratemporal condition:

$$p_t^N = \frac{1-\omega}{\omega} \left(\frac{x_t^T}{x_t^N} \right)^{\frac{1}{\xi}}, \quad (10)$$

which pins down the relative price of non-tradables. Keeping x_t^N fixed, p_t^N is increasing in x_t^T with constant elasticity determined by ξ . Intuitively, the more complementary both goods are, which translates to lower ξ , the higher the price response to changes in the use of tradable inputs will be.

Final Tradable Goods Producers. A competitive firm aggregates differentiated varieties $y_t^T(i)$ to produce a final tradable good according to:

$$y_t^T = \left[\int_0^1 y_t^T(i)^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}} \quad (11)$$

Solving their cost minimization problem yields a demand for variety i

$$p(i) = \left(\frac{y_t^T}{y_t^T(i)} \right)^{\frac{1}{\eta}}, \quad (12)$$

which depends on total production y_t^T and the price of variety i , $p(i)$. The elasticity of substitution between varieties, η , determines the elasticity of demand.

Intermediate Goods Firms. Firm i uses capital and labor h to produce its variety according to the technology

$$y(i) = A(i)k(i)^\alpha h(i)^{1-\alpha} \quad (13)$$

Productivity is determined by an aggregate component A and an invariant firm-specific component $a(i)$ such that, in logs,

$$\log A(i) = \log A + \log a(i), \quad (14)$$

where, across firms, $\log a(i) \sim \mathcal{N}(0, \sigma_a^2)$. While all firms face the same wage w , I assume that they face different costs of capital $R(i)$, which for now I take as given. Internalizing demand for their variety, their problem is given by

$$\max_{k_t(i), h_t(i)} (1+s) \left(A(i)k(i)^\alpha h(i)^{1-\alpha} \right)^{\frac{\eta-1}{\eta}} y_t^{\frac{1}{\eta}} - R_t(i)k_t(i) - w_t h_t(i), \quad (15)$$

where $1+s = \frac{\eta}{\eta-1}$ is a production subsidy financed with a lump-sum tax that corrects the inefficiency arising from the markup.⁹

The following standard result from the misallocation literature allows me to aggregate firms into a representative firm.

Lemma 1 (Firm Aggregation). *Let σ_2^R be the variance of the cost of capital, $R(i)$. Up to second order, tradable production can be represented by an aggregate firm with production technology*

$$y_t^T = \text{TFP}_t k_t^\alpha h_t^{1-\alpha}, \quad (16)$$

where

$$\log \text{TFP}_t = \log A_t + \frac{1}{2}(\eta-1)\sigma_a^2 - \frac{1}{2}\alpha(1+\alpha(\eta-1))\sigma_R^2 \quad (17)$$

and the solution to its problem is characterized by,

$$w_t = (1-\alpha) \text{TFP}_t \left(\frac{k_t}{h_t} \right)^\alpha \quad (18)$$

$$R_t = \alpha \text{TFP}_t \left(\frac{h_t}{k_t} \right)^{1-\alpha} \quad (19)$$

This approximation is exact if $R(i)$ is log-normally distributed.

Proof. See [A.1](#) □

⁹This subsidy, which is analogous to the commonly used subsidy in New-Keynesian models, ensures that production is at its socially optimal level. Removing this distortion simplifies the analysis of optimal policy by reducing the scope of intervention for the planner with no qualitative implications.

This result is very useful to keep the analysis tractable because it shows that it is not necessary to keep track of the entire distribution of $R(i)$, just its variance σ_R^2 . As a corollary, it follows that for any variable to have an impact on productivity, it must affect the variance of $R(i)$.

Market Clearing. Demand for non-tradables must equal its endowment:

$$x_t^N = y^N \quad (20)$$

In the same way, the consumption of final goods must equal their production:

$$c_t + k_{t+1} - (1 - \delta)k_t = x_t \quad \forall t \quad (21)$$

Similarly, the factor markets clear:

$$h_t = 1 \quad \forall t \quad (22)$$

$$k_t = k_t \quad \forall t \quad (23)$$

I now explain in detail the determination of σ_R^2 , which will lead to the policy trade-off.

2.2 Capital Misallocation

In this section I introduce the key departure in this model from the existing small open economy models in order to model σ_R^2 and its dynamics.

Capital is owned by households, who decide on $t - 1$ how much capital to supply in t . Domestic banks intermediate between the firms and the households. Credit markets are segmented, with each market corresponding to a single firm and bank.¹⁰ Each bank takes deposits $\hat{k}(i)$ from the households, which they use to supply the firm in their market and cover intermediation costs, given by the realization of $f(i)$. As such, capital available to firm i is given by

$$k_t(i) = \hat{k}_t(i) - f(i)$$

where $f(i)$ stands in for both overhead costs such as setting up a branch or a relationship with a firm as well as variable costs.¹¹

¹⁰For simplicity, I assume that these banks are perfectly competitive so that there's no interest rate spread. As with the assumption on the subsidy that eliminates the intermediate good markup, I introduce this solely to reduce the number of dimensions on which the competitive equilibrium differs from the efficient allocation.

¹¹ $f(i)$ could also stand for unexplained credit demand such as government borrowing.

Given market clearing, this determines the cost of capital faced by the firm, and charged by the bank, which is described by

$$R(i) \propto \left(\hat{k}_t(i) - f(i) \right)^{-\frac{1}{1+\alpha(\eta-1)}}$$

Banks choose $\hat{k}_t(i)$ to maximize their expected profits $R(i) \left(\hat{k}_t(i) - f(i) \right) - R\hat{k}_t(i)$, where R is the rate paid to households. To keep this problem tractable, I make the following two assumptions:

Assumption 1 (Segmented Markets). *After shocks $f(i)$ are realized, banks cannot transfer capital between markets.*

Assumption 2 (Shock Scaling). *Shocks are given by $f(i) = A(i)^{\nu(\eta-1)}\hat{k}(i)^{1-\nu}F(i)$, where $F(i)$ is a random variable and $\nu \geq 0$.*

Assumption 2 provides structure on the dependence on the intermediation costs on $\hat{k}(i)$, which is given by ν . For $\nu < 1$, costs scale less than one by one with supplied capital, which would occur if a part of them are fixed. For $\nu = 1$, costs are fixed and do not depend on intermediated capital. Lastly, $\nu > 1$ would imply that total intermediation costs decrease as capital supply increases. In this paper, I will focus on the case where $\nu \leq 1$.

The bank's problem is to choose the amount of deposits $\hat{k}(i)$ from households. Assumption 1 and 2 ensure that the problem will be symmetric across banks up to the productivity of the firm, as shown by the following Lemma.

Lemma 2. *If costs $F(i)$ are scaled by $A(i)^{\nu(\eta-1)}$, then, in equilibrium, each individual bank takes deposits*

$$\hat{k}(i) = \frac{A(i)^{\eta-1}}{\mathbb{E}[A(i)]^{\eta-1}} k \quad (24)$$

Proof. See [A.2](#) □

As a result, the market clearing rental rate of capital $R(i)$ faced by firm i is

$$R(i) \propto \left(k_t \left(1 - k_t^{-\nu} F(i) \right) \right)^{-\frac{1}{1+\alpha(\eta-1)}} \quad (25)$$

In broad terms, this formulation captures two empirical regularities: first, firms with the same fundamentals use different amounts of capital; second, these distortions are decreasing in the aggregate amount of capital in the economy. I now go over these two points in more detail.

Firms with positive (negative) draws of $F(i)$ will use more (less) capital and pay lower (higher) rental rates. It follows that the larger the variance of $F(i)$ is, the larger σ_R^2 will be, as the following Lemma shows.

Lemma 3 (Rental Rates Dispersion). *For small distortions $F(i)$, the variance of $\log R(i)$, σ_R^2 , is given by*

$$\sigma_R^2 = \left(\frac{1}{1 + \alpha(\eta - 1)} \frac{1}{k_t^\nu} \right)^2 \text{Var} [F(i)], \quad (26)$$

Proof. See [A.3](#) □

This closed form solution is an important feature, as it makes very clear the link between the aggregate capital stock and σ_R^2 and allows for an analytical characterization of the elasticity of TFP to aggregate capital, defined as

$$\gamma_t \equiv \frac{\partial \log \text{TFP}_t}{\partial \log k_t}$$

Combining Lemmas [1](#) and [3](#), it is easy to see that

$$\gamma_t = \alpha(1 + \alpha(\eta - 1))\nu \sigma_R^2, \quad (27)$$

which captures how an increase in the amount of capital improves its allocation across firms. γ_t is increasing in both the initial level of misallocation in the economy, given by σ_R^2 , and its elasticity with respect to k_t , ν . The latter captures how the frictions in the economy, given by the intermediation costs $F(i)$, scale with k_t . For low values of ν , these frictions are almost proportional to k_t , which means that increases in investment do not lead to significant improvements in capital allocation. On the other hand, if ν is high, small increases in k_t greatly decrease misallocation.

Discussion of Formulation. Equation [\(25\)](#) introduces capital misallocation as the result of reduced form intermediation costs. This has the main benefit of being very tractable, as Lemma [3](#) shows, a feature I will leverage in the quantification section of this article. I argue that this specification is also in line with both empirical facts and the theoretical literature on misallocation.

Regarding the empirical facts, this specification has two main predictions. First, the model predicts that economies with a higher investment rate will experience higher productivity growth. This is consistent with the literature on the deepness of financial markets and growth (Rajan and Zingales, [1998](#); Levine, [2005](#)).

Second, misallocation of capital across firms increases (decreases) with decreases (increases) in credit. This is line with the literature that looked at the effect of capital controls (Bau and Matray, 2023; Larrain and Stumpner, 2017; Varela, 2017), macroprudential policies (Jiménez et al., 2017; Blattner, Farinha, and Rebelo, 2023), and government spending (Pinardon-Touati, 2024). This qualitative implications are also the same as in the models used by Larrain and Stumpner (2017) and Varela (2017) to interpret their findings.

On the theoretical side, I show in Online Appendix B.1 that equation (25) can be derived from two alternative models that embody commonly used assumptions when studying misallocation: borrowing constraints arising from limited commitment, and heterogeneity in access to equity across firms. In this way, the simplicity of the model is also a feature, as it accommodates a wide class of frictions that can lead to misallocation of capital.

While firms in this model only face one distortion, I show in Online Appendix B.2 that the model can accommodate other distortions, and show that γ retains a closed form expression when including labor misallocation.

3 Competitive Equilibrium

Having described the environment, I now define the competitive equilibrium of this economy:

Definition (Competitive Equilibrium). *A competitive equilibrium is a set of (state-contingent)*

1. *household allocations* $\{c_t, d_{t+1}, k_{t+1}\}$
2. *firm allocations* $\{x_t, x_t^T, x_t^N, k_t, h_t\}$
3. *rental rates dispersion* σ_R^2
4. *prices* $\{w_t, p_t, p_t^N, R_t\}$
5. *Government policy* $\{\tau_t^d, \tau_t^k, T_t\}$

such that, given initial allocations $\{k_0, d_0\}_t$, the non-tradable endowment y^N , world interest rate r , A_t , $\bar{d}_t$, and σ_a^2 ,

1. *Given prices and government policy, the allocations solve the household problem.*
2. *Given prices, the allocations solve the firm's problem.*
3. *The rental rate dispersion is consistent with profit maximizing by firms.*
4. *Markets clear*
5. *The government budget is balanced*

3.1 Characterization

Factor prices are pinned down by the combination of Lemma 1 and market clearing,

$$w_t = (1 - \alpha) \text{TFP}_t k_t^\alpha \quad (28)$$

$$R_t = \alpha \text{TFP}_t k_t^{\alpha-1} \quad (29)$$

Combining factor prices, the budget constraint of the household, non-tradable market clearing, the government's budget constraint and the firm's problem yields the tradable resource constraint,

$$x_t^T = \text{TFP}_t k_t^\alpha - (1 + r)d_t + d_{t+1} \quad (30)$$

A resource constraint for final goods can be derived from market clearing and firms conditions

$$c_t + k_{t+1} = (1 - \delta)k_t + \left[\omega x_{T,t}^{\frac{\xi-1}{\xi}} + (1 - \omega)y^N \frac{\xi-1}{\xi} \right]^{\frac{\xi}{\xi-1}} \quad (31)$$

Market clearing and (9) pin down the price of final goods:

$$p_t = \frac{1}{\omega} \left(\frac{c_t + k_{t+1} - (1 - \delta)k_t}{x_t^T} \right)^{\frac{1}{\xi}} \quad (32)$$

Combining market clearing and the intratemporal condition (10) yields an expression for the relative price of non-tradable goods,

$$p_t^N = \frac{1 - \omega}{\omega} \left(\frac{x_t^T}{y^N} \right)^{\frac{1}{\xi}} \quad (33)$$

Because both goods are complements, an increase in tradable consumption pushes up demand for non-tradable goods. Given that supply is perfectly inelastic, this leads to an increase in the price p_t . Because this is a market clearing condition, it depends on aggregate tradable consumption x_t^T and households do not internalize this effect of their consumption choices.

To obtain a borrowing constraint that depends on allocations, I use (28), (29) and (33) in (3), which yields

$$d_{t+1} \leq \kappa \left(\text{TFP}_t k_t^\alpha + \frac{1 - \omega}{\omega} \left(\frac{x_t^T}{y^N} \right)^{\frac{1}{\xi}} y^N \right) + \bar{d}_t. \quad (34)$$

Households can borrow up to a fraction κ of their income, which is the sum of tradable output and the value in tradable goods of their non-tradable endowment. Aggregate tradable

consumption enters this constraint through its effect on the price of non-tradable goods.

I re-state the Euler equation (4) for debt, which remains unchanged, for convenience below,

$$\frac{c_t^{-\sigma}}{p_t} = \frac{1}{1 - \mu_t} \beta(1 + r)(1 + \tau_t^d) \mathbb{E} \left[\frac{c_{T,t+1}^{-\sigma}}{p_{t+1}} \right],$$

Combining the Euler equation for capital (6) and (29) yields

$$c_t^{-\sigma} = \beta \mathbb{E} \left[c_{t+1}^{-\sigma} \left(\frac{\alpha \text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{p_{t+1}} (1 + \kappa \mu_{t+1}) + \frac{\tau_{t+1}^k}{p_{t+1}} + 1 - \delta \right) \right] \quad (35)$$

Lastly, given Lemmas 1-3, TFP_t can be written as

$$\text{TFP}_t = \text{TFP}(A_t, k_t) \quad (36)$$

3.2 Sources of Inefficiency

Before moving on to analyzing optimal policy, I discuss the two features of the model that make this equilibrium inefficient. First, as it is common in this class of models, the presence of the price of non-tradable goods in the borrowing constraint (34) introduces a pecuniary externality. This is because p_t^N depends on aggregate tradable consumption, an effect not internalized by households. Suppose that the economy hits the borrowing constraint in period t . Because households are not able to borrow as much as desired, they reduce tradable consumption, as shown by condition (4) when $\mu_t > 0$. Lower tradable consumption decreases the value of the non-tradable endowment, lowering the maximum amount households can borrow.

A reduction in d_{t+1} reduces resources available to households in (30), further reducing consumption and investment. As I will discuss in the next section, a planner that internalizes this will place a higher value on consumption relative to households in periods where the borrowing constraint binds. In this setting, macroprudential policy can be welfare improving by increasing consumption during these episodes, which is achieved by reduced borrowing in previous periods.

The second source of inefficiency is the misallocation of capital across firms, which reduces the productivity of the economy, as indicated by Lemma 1. Ideally, the government would like to set the distortions to zero, increasing productivity. Even if it cannot do so, as I will assume, there's still scope for intervention. As shown by Lemma 3, an increase in investment reduces the weight of these distortions, leading to a better allocation of capital across firms and a lower σ_R^2 . The sufficient statistic for this channel will be the elasticity of productivity

to capital, γ_t .

While the pecuniary externality calls for restrictions on consumption, the imperfections on the capital market mean that increases in investment can be beneficial. Interestingly, both have opposite effects on borrowing. Given sufficient policy instruments, this poses no problem.¹² If the government, however, is limited to controlling only gross borrowing, then a conflict between objectives arises.

In the following section, I discuss optimal policy and this potential trade-off in more detail.

4 Optimal Policy

In this section I characterize how a benevolent planner can address the inefficiencies in this economy. I am interested in the problem of a government that internalizes the effect of aggregates but is still subject to the borrowing constraint and cannot control the allocation of capital between firms.¹³ I start by considering the solution to this problem when the government can set both the capital control τ_t^d and the investment subsidy τ_t^k , which I call the constrained efficient allocation. This is a useful benchmark and is illustrative of the mechanisms at play.

4.1 Constrained Efficient Allocation

I define this allocation as the solution to the problem of a planner that chooses allocations $\{x_t^T, d_{t+1}, k_{t+1}\}$ subject to resource constraints (30,31), the borrowing constraint (34) and the process for TFP. Including the latter assumes that the policymaker has no ability to modify the allocation of capital across firms.

Along with the constraints, the solution is characterized by,

$$\frac{c_t^{-\sigma}}{p_t} = \frac{1}{1-\mu} \beta(1+r) \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \left(1 + \kappa \mu_{t+1} \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right) \right], \quad (37)$$

$$c_t^{-\sigma} = \beta \mathbb{E}_t \left[c_{t+1}^{-\sigma} \left((\alpha + \gamma_{t+1}) \frac{\text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{p_{t+1}} \right) \left(1 + \kappa \mu_{t+1} \left(1 + \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right) \right) \right. \\ \left. + (1-\delta) \right], \quad (38)$$

¹²The open economy assumption does some implicit lifting here. In a closed economy, savings equal investment and so it would not be possible to target both simultaneously. In an open economy, the difference between savings and investment is the current account, which gives the extra degree of freedom to the policymaker.

¹³In other words, the government takes the financial system as a technological constraint.

where p_t and p_{t+1}^N are given by (9) and (10) respectively.

Condition (37) pins down optimal borrowing. Compared to the household's condition (4), note that the planner internalizes the effect of its choices on prices when there's a sudden stop. An extra dollar of debt in t reduces available resources in $t + 1$ by $1 + r$. In turn, the share of non-tradable to tradable consumption times the price elasticity predicts the resulting drop in the value of the non-tradable endowment. Lastly, any drops in the value of income reduces borrowing by κ .

The choice of capital is pinned down by (38). Comparing to its competitive equilibrium analog, (35), the planner internalizes two external effects of investment. First, a higher stock of capital mitigates misallocation across firms, raising TFP_{t+1} . This is shown by the presence of γ_{t+1} , which measures the improvement in productivity that derives from increased investment.

Second, by increasing output in $t + 1$, investment increases available resources to the household. This will, ceteris paribus, increase consumption by the marginal productivity of capital. In turn, this increases the value of the non-tradable endowment and the borrowing capacity of the households.

What are the policy implications of this analysis? Proposition 1 derives the optimal taxes as a function of key sufficient statistics that summarize the channels previously discussed.

Proposition 1 (Constrained Efficient Allocation Implementation). *The constrained efficient allocation can be implemented as a competitive equilibrium with a tax on debt τ_t^d and a subsidy on investment τ_t^k that take the following values,*

$$\tau_t^d = \mathbb{E} \left[\frac{c_{T,t+1}^{-\sigma}}{p_{t+1}} \right]^{-1} \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \kappa \mu_{t+1} \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right] \quad (39)$$

$$\tau_{t+1}^k = \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right]^{-1} \mathbb{E}_t \left[c_{t+1}^{-\sigma} \left(\frac{\text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{p_{t+1}} \right) \left(\gamma_{t+1} (1 + \kappa \mu_{t+1}) + (\alpha + \gamma_{t+1}) \kappa \mu_{t+1} \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right) \right] \quad (40)$$

Proof. See A.4 □

The tax on debt, τ_t^d , captures the difference between the social and the private value of a consumption. As previously discussed, because households do not internalize the aggregate effects of their choices, they consume less than it is socially optimal during a sudden stop. The tax then increases the cost of consumption in t so that households internalize this. Note that the tax also depends on the severity and likelihood of a crisis, as reflected by the expectation of the multiplier on the constraint. If $\mu_{t+1} = 0$ for all possible states in $t + 1$,

then the social and private values of consumption are aligned. Likewise, if the sudden stops do not feature large drops in consumption, it follows from (5) that μ_{t+1} will be small, pushing down the tax on debt.

The investment subsidy, τ_t^k , reflects the two sources of inefficiency in this model. The first term captures the additional social value of investment over its private returns, due to its positive effects on productivity, reflected by γ_{t+1} . Additional investment also increases the resources available to the household, which increases consumption and thus borrowing capacity. This is captured by the second term.

Because it has two available instruments, the policymaker faces no trade-off in its objectives. This is clear in Proposition 1; The capital control τ_t^d does not depend on the productivity externality γ_{t+1} . Likewise, the presence of sudden stops does not reduce the investment subsidy.¹⁴ Thus, in contrast with the literature, this economy does not feature overborrowing in the proper sense of the word; rather, it features overconsumption and underinvestment.

4.2 Second-Best Capital Controls

The previous result depends on the ability of the government to control the composition of credit between consumption and investment. In practice, however, governments may have trouble setting restrictions on each kind of credit. In this section, I present one of the main results of the paper: Governments that can only control aggregate borrowing face a trade-off between financial stability and productivity growth.

This constrained government can set capital controls but cannot subsidy investment. Formally, this is equivalent to adding a constraint that $\tau^k = 0$ to the planner's problem. It is clear from inspecting the optimality conditions of the household (4)-(35), that the planner cannot independently implement k_{t+1} and c_t anymore. Reducing consumption requires increasing τ_t^d to raise the interest rate paid by the households, which disincentivizes investment. To increase investment, the interest rate must decrease, which implies lowering τ_t^d . In turn, this will lead to an increase in consumption.

It is possible to use this intuition to set up a tractable version of this problem¹⁵. I assume that households are not able to borrow and instead receive transfers from the government, who borrows on their behalf. The solution to this problem yields policy functions for c_t and k_{t+1} that depend on d_{t+1} . The key object is the response of investment to increases in d_{t+1} ,

¹⁴In fact, the subsidy is increasing in μ_{t+1} .

¹⁵As opposed to setting up the Ramsey problem with the household optimality conditions as implementability constraints

which I call the marginal propensity to invest, mpi, given by

$$\text{mpi} \equiv \frac{\partial k_{t+1}}{\partial d_{t+1}} \quad (41)$$

The planner solves for the optimal d_{t+1} , taking as given the policy functions for c_{t+1} and k_{t+1} , and implements it using τ^d . Given that in this small open economy d_{t+1} has a close link to the current account, it is also a natural extension when studying capital controls.¹⁶

After application of the envelope theorem, the first order condition of the planner is given by

$$\begin{aligned} \frac{c_t^{-\sigma}}{p_t} = \beta \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} (1 + r \right. \\ \left. + \kappa \mu_{t+1} ((1 + r) - \text{mpi} \times ((\alpha + \gamma_{t+1}) \text{TFP}_{t+1} k_{t+1}^{\alpha-1})) \times \frac{1}{\xi} \frac{p_{t+1} y^N}{x_{t+1}^T} \right. \\ \left. - \text{mpi} \times \gamma_{t+1} \text{TFP}_{t+1} k_{t+1}^{\alpha-1} (1 + \kappa \mu_{t+1})) \right] \end{aligned} \quad (42)$$

An extra dollar of borrowing increases the resources available to households, the benefit of which is measured by the marginal utility of income. This increase in debt also reduces resources in $t + 1$ by $(1 + r)$, which is captured by the first term on the right-hand side. The extra terms correspond to the external effects of household choices.

As previously discussed, by reducing resources available for consumption, borrowing has an additional negative effect during a sudden stop. Unlike in the constrained efficient problem, here borrowing also has effects on investment. An extra dollar of debt leads to a mpi increase in k_{t+1} . Because investment increases available resources, this mitigates the cost of borrowing before a sudden stop. The other external effect of additional borrowing is the productivity increase that follows from investment. The strength of this effect will also depend on how the economy allocates credit, mpi, and the elasticity of TFP_{t+1} to investment, γ_{t+1} .

These two effects have opposing implications for welfare, implying a policy trade-off. Increases in borrowing allow for additional investment, which has positive effects on productivity. At the same time, this increased leverage is costly during a sudden stop. The following proposition shows the second-best capital control, $\tau_t^{d, sb}$, balances this trade-off.

Proposition 2 (Implementing the Second-Best Allocation). *The tax on debt $\tau_t^{d, sb}$ that implements the second-best allocation is*

$$\tau_t^{d, sb} = \tau_t^d - \frac{\text{mpi}}{1 + r} \times \tau_{t+1}^k, \quad (43)$$

¹⁶Re-arrange the resource constraint to obtain $-(d_{t+1} - d_t) = (\text{TFP}_t k_t^\alpha - x_t^T) - r d_t$. As first term on the RHS is the trade balance, the RHS is the current account ca_t . As a result, $ca_t = -(d_{t+1} - d_t)$.

where τ_t^d and τ_{t+1}^k , defined in Proposition 1, are the instruments that implement the constrained efficient allocation.

Proof. See A.5 □

Proposition 2 shows that the constrained planner sets a capital control that is a linear combination of the constrained efficient instruments. The weight on the investment subsidy τ^k is given by the marginal propensity to invest, mpi, which determines the response of investment to foreign borrowing.

5 Determinants of Optimal Policy

In this section I explore the main determinants of capital controls. To fix ideas, I focus on each source of inefficiency to study how it affects optimal capital controls.

5.1 Pecuniary Externality

Perfect foresight and no investment. To help with the intuition, I first consider a case where the borrowing constraint binds with probability equal to 1 in the next period, there are no other sources of uncertainty and tradable production is an endowment $y_{t+1}^T = y^T$. Under this scenario, the optimal tax is

$$\tau_t^{d, sb} = \tau_t^d = \kappa \mu_{t+1} \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T}$$

Without investment, this solution matches the constrained efficient allocation, as there's no trade-off for the planner. As all terms are strictly positive, this implies a positive tax on debt that reduces borrowing and, as a result, tradable consumption. Moreover, the tax is increasing in both of the terms.

The second term gives the response of the value of the non-tradable endowment to changes in x_{t+1}^T . It depends on the price elasticity $\frac{1}{\xi}$, which is inversely related to the substitutability between goods, and the share of non-tradable to tradable consumption. The higher this response is, the more effective macroprudential will be, as any increase in x_{t+1}^T will lead to larger increases in the value of the non-tradable endowment. The amount of income that can be pledged as collateral, given by κ , has a similar effect, as it translates the changes in income to changes in borrowing.

The tax also depends on the severity of the sudden stop, which is given by μ_{t+1} . Recall from (5) that μ_{t+1} reflects the consumption drop during a sudden stop. It follows that

capital controls should be higher for countries where crises are more harmful in terms of consumption.

Perfect foresight and investment. How does the introduction of investment affect the results? I now bring back tradable production with exogenous productivity (i.e. $\gamma_{t+1} = 0$), so that the pecuniary externality remains the only source of inefficiency. The second-best capital control is now given by

$$\tau_t^{d, sb} = \kappa \mu_{t+1} \left(1 - \text{mpi} \times \frac{\alpha \text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{1+r} \right) \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T}$$

The introduction of capital means that households now allocate resources between consumption and investment and these choices are an important input for policy. Note that, an extra dollar in borrowing during t leads to mpi additional cents in investment. This has the effect of increasing production, and thus available resources, during $t+1$. Because macroprudential policy seeks to control consumption, it is possible to think of the mpi as measuring an additional cost of capital controls. Thus, the higher the propensity to invest is, the lower the tax will be.

Within the model studied in this paper, mpi is characterized by the optimality conditions and is therefore implicitly determined by the primitives of the model. Framing the discussion in terms of the former, however, allows for more generality. Indeed, one could think of more complex settings, such as models with rich heterogeneity within households, a more detailed production structure or financial system, in which this sufficient statistics still applies.

5.2 Misallocation

Perfect foresight and no borrowing constraint. As previously discussed, the frictions that lead to misallocation result in a positive externality of investment, as a higher capital stock increases total factor productivity. To make the intuition clear, I will keep the perfect foresight assumption for the time being while assuming that there's no borrowing constraint (i.e., $\mu_{t+1} = 0$). In that economy, the second-best capital control would be

$$\tau_t^{d, sb} = -\text{mpi} \times \gamma_{t+1} \times \frac{\text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{1+r}$$

The second term is the constrained efficient subsidy, that depends on γ_{t+1} . Recall from (27), that γ_{t+1} depends in part on the existing level of misallocation in the economy. This has two important implications; the first is that the subsidy will vary across countries. It is an

empirical regularity¹⁷ that misallocation is decreasing in the level of development, implying that the subsidy would be higher for less developed economies. The second is that within a country, as the capital stock increases, misallocation will fall and γ_{t+1} will decrease. In this way, the externality resembles a dynamic externality that shrinks as the economy develops (Redding, 1999; Melitz, 2005; Itskhoki and Moll, 2019; Ottonello, Perez, and Witheridge, 2024).

The size of the subsidy also depends on the efficiency of the government in increasing investment, given by mpi . If the economy mostly redirects borrowing to consumption, that would make the subsidy relatively more expensive, and its optimal level would be higher. An interesting implication is that the subsidy will more effective for economies that either have a naturally higher mpi or can control it in some way. This is consistent, for instance, with the experience of South Korea (Noland, 2007).

So far, I have considered a perfect foresight scenario, where the sudden stop is all but certain. In reality, sudden stops are low probability events¹⁸. This is important because it introduces a third factor into the trade-off which has asymmetric effects on the two forces at play. While increases in productivity will occur in all states of the world, the borrowing constraint only binds in a few of them.

Uncertainty. First, assume that $\gamma_{t+1} = 0$. Then, the second-best capital control is

$$\tau_t^{d, sb} = \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right]^{-1} \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \kappa \mu_{t+1} \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right]$$

To unpack the role the probability of a sudden stop plays, let π be the probability that $\mu_{t+1} > 0$. It is possible to write

$$\tau_t^{d, sb} = \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right]^{-1} \times \pi \times \mathbb{E} \left[\left(\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right) \kappa \mu_{t+1} \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \middle| \mu_{t+1} > 0 \right].$$

As long as $\pi < 1$, the introduction of uncertainty reduces the optimal capital control. This is intuitive, because its benefits only manifest during a sudden stop, while its costs occur in every state of the world.

These implications also carry over to the case where $\gamma_{t+1} > 0$. Because the benefits to productivity happen in all states of the world, a lower probability of a sudden stop reduces the benefit to capital controls while keeping their costs intact.

¹⁷(See, e.g., Hsieh and Klenow, 2009; Buera, Kaboski, and Shin, 2015; Restuccia and Rogerson, 2017)

¹⁸Bianchi and Mendoza (2020) report a frequency of 2.4% with significant heterogeneity across development levels

What are the overall implications of each of the factors discussed above? The following corollary summarizes the contribution of each.

Corollary. *The second-best capital controls, $\tau_t^{d, sb}$ are*

1. *Increasing in the non-tradable share of consumption $\frac{p_{t+1}^N y^N}{x_{t+1}^T}$, non-tradable price elasticity, $\frac{1}{\xi}$, pledgeable fraction of income κ , sudden stop severity, μ_{t+1} and sudden stop probability, π*
2. *Decreasing in misallocation, σ_R^2 , its elasticity with respect to capital, ν , and the marginal propensity to invest in tradables mpi*

The benefits of capital controls are predominantly given by the strength of the pecuniary externality, which leads the planner to restrict consumption using capital controls. The introduction of investment reduces the effectiveness of capital controls because restricting borrowing leads to decreases in investment, which are costly. The marginal to propensity to invest, mpi , summarizes this channel.

While investment dampens optimal capital controls, the presence of frictions in the allocation of capital further increase the costs of capital controls. Because these frictions diminish as the stock of capital increases, capital controls reduce productivity. The strength of this channel is summarized by γ_{t+1} , which in turn depends on the level of misallocation in the economy, σ_R^2 and the elasticity of misallocation with respect to k_{t+1} , ν .

6 Quantification

In this section, I leverage the results of Proposition 2 to estimate the second-best capital controls and show the quantitative relevance of the channels discussed in the previous section. While Proposition 2 states the second-best capital controls as a function of a few sufficient statistics and time-series moments, a few assumptions are needed before taking it to the data. I discuss these before describing the empirical work.

Building on the intuition from the previous section, it is possible to write τ_d and τ^k as

$$\begin{aligned}\tau^d &= \pi \times \kappa \frac{1}{\xi} \times \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \right]^{-1} \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \mu_{t+1} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \middle| \mu_{t+1} > 0 \right] \\ \tau^k &= \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \right]^{-1} \left[\gamma_{t+1} \times \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \frac{y_{t+1}^T}{k_{t+1}} \right] + \right. \\ &\quad \left. \pi \times \kappa \times \mathbb{E} \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \left(\mu_{t+1} \left(\gamma_{t+1} \frac{y_{t+1}^T}{k_{t+1}} + \left((\alpha + \gamma_{t+1}) \frac{y_{t+1}^T}{k_{t+1}} \right) \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right) \right) \middle| \mu_{t+1} > 0 \right] \right]\end{aligned}$$

where π is the probability of a sudden stop in $t + 1$. To simplify the estimation I assume that, conditional on a sudden stop, $\frac{p_{t+1}^N y^N}{x_{t+1}^T}$ and TFP_{t+1} are constant.

I now explain how I assign values to the parameters and moments that are required.

6.1 Calibrated Parameters, Sufficient Statistics and Time-Series Moments

Table 1: Baseline Calibration of Standard Parameters

Parameter	Value	Source
σ	2	Bianchi (2011)
ξ	0.83	Bianchi (2011)
κ	0.32	Bianchi (2011)
r	0.04	Bianchi (2011)
α	0.28	Standard Value
δ	0.1	Standard value
η	3	Hsieh and Klenow (2009)

Calibrated Parameters. Table 1 summarizes the calibration of the standard parameters. I keep as close as possible to Bianchi (2011) in calibrating the parameters that pin down the pecuniary elasticity and set $\sigma = 2, \kappa = 0.32$ and $\xi = 0.83$. I also follow Bianchi (2011) in setting $r = 0.04$, as it is common in the literature.

For the parameters governing tradable production, I set $\alpha = 0.28$ and $\delta = 0.1$ as is standard. The elasticity of substitution between varieties, η , determines the effect of misallocation on productivity, as can be seen in Lemma 1. I follow Hsieh and Klenow (2009) and make the conservative choice of $\eta = 3$.

Calibrated Sufficient Statistics. Table 2 lists the sufficient statistics that I calibrate following the literature. To measure the marginal propensity to invest, I set $\text{mpi} = 0.2$ as a conservative value between Müller and Verner (2023) who report that, during credit booms, credit to the tradable sector is around 10% of the total expansion, and Hong (2023), who estimate a marginal propensity to save of almost 40 for emerging economies.

I follow two approaches to assign values to π . First, I use the unconditional probabilities described by Bianchi and Mendoza (2020), which are 2.4% for their entire sample, and 2.9%

and 1.7% for emerging and advanced economies respectively. Second, a recent literature on financial crises has tried to find predictors of future crises. Greenwood et al. (2022) construct “Red-Zone” indicators, which combines several financial series. Countries in red-zones have a probability of around 14% of experiencing a crisis the next year and between 37-45% within the next three years.

Lastly, I need to assign a value to the ratio of non-tradable to tradable expenditure $\frac{p_{t+1}^N y^N}{x_{t+1}^T}$, which affects the response of p_{t+1}^N to changes in x_{t+1}^T and is therefore important in determining the importance of the pecuniary externality. I follow Bianchi and Mendoza (2020) and set $\frac{p_{t+1}^N y^N}{x_{t+1}^T} = 2$.

Table 2: Calibrated Sufficient Statistics

Sufficient Statistic	Value	Source
mpi	0.2	Middle point between Müller and Verner (2023) and Hong (2023)
π	0.017 - 0.45	Bianchi and Mendoza (2020) and Greenwood et al. (2022)
$\frac{p_{t+1}^N y^N}{x_{t+1}^T}$	2	Bianchi and Mendoza (2020)

Time-Series Moments. I combine data from the World Development Indicators (World Bank, 2021) and Penn World Tables (Feenstra, Inklaar, and Timmer, 2015) to obtain a panel containing national accounts data and capital stock. Following Uribe and Schmitt-Grohé (2017), I keep only those countries with at least 30 years of available data, and quadratically detrend all series. To identify sudden stops, I use Korinek and Mendoza (2014) classification. The resulting panel spans 36 countries from 1979 to 2012, and covers middle and high-income economies.¹⁹²⁰

Table 3 shows the moments required for the estimation. To estimate μ_{t+1} , I use (5) and proxy for $\mu_{t+1} > 0$ using the sudden stop classification in Korinek and Mendoza (2014). Lastly, I obtain the output to capital ratio using data from Feenstra, Inklaar, and Timmer (2015).

¹⁹I assign to each country its modal WDI classification over the period 1989, when the WDI classification started, and 2012.

²⁰Section C.1 gives some additional details.

Table 3: Time-Series Moments

Moment	Value
$\mathbb{E} \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \right]$	1
$\mathbb{E} \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \mu_{t+1} \mu_{t+1} > 0 \right]$	0.05
$\mathbb{E} \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{p_t}{p_{t+1}} \text{TFP}_{t+1} k_{t+1}^{\alpha-1} \right]$	0.25
$\text{TFP}_{t+1} k_{t+1}^{\alpha-1}$	0.25

6.2 Productivity Externality

The key sufficient statistic I need to estimate is γ_{t+1} . Recall from (27) that it is given by,

$$\gamma_t = \alpha(1 + \alpha(\eta - 1))\nu \sigma_R^2,$$

which requires values for σ_R^2 and ν .

To estimate σ_R^2 , I follow the procedure used by Hsieh and Klenow (2009). Recall that σ_R^2 is the variance of the log of the rental rates, $R(i)$, paid by the firms producing tradable goods varieties. It follows from profit maximization that

$$R(i) = \alpha \frac{p(i)y^T(i)}{k(i)}, \quad (44)$$

Where the denominator is the value added of the firm and the numerator the stock of capital it possesses. Thus, estimating σ_R^2 requires firm-level data that allows to estimate the ratio of these two numbers. The Orbis-Amadeus historical product, which has extensive coverage of private firms in Europe (Kalemli-Özcan et al., 2024), provides exactly that. I now detail the steps I take to measure σ_R^2 .

Data Cleaning. To clean the dataset I follow the procedure by Kalemli-Özcan et al. (2024) as closely as possible. The input for the Orbis dataset is income and balance sheet statements submitted annually by firms. To have consistent units, I keep only unconsolidated statements that cover 12 months.

I drop spells with errors in the following way. First, I tag unrealistic changes in assets or sales²¹, or negative values for assets, sales, employment or liabilities. I also tag observations

²¹In the order of 10^3

that do not report employment or report a number larger than 2 million, or where balance sheet identities don't hold. I drop all tagged observations and only keep the spell after the last identified error.

As it is common in the literature that measures misallocation, I focus on the manufacturing sector²², defined using the 4-digit NACE classification. Section C.2 presents some summary statistics of the resulting sample which spans 18 countries²³ over the period 1996-2016, covering 1,050,610 unique firms for a total of 9,143,358 observations. Employment by these firms adds up to 11,559,081 employees per year, with the firm size distribution resembling that reported by Eurostat.²⁴

Variable Construction. I construct $p(i)y^T(i)$ by subtracting the cost of materials from the operating revenue of the firm and the capital stock $k(i)$ as the sum of tangible and intangible fixed assets. To mitigate potential measurement error, I winsorize both variables at the bottom and top 1% levels. I then estimate $R(i)$ using equation (44) and its log variance at the 4-digit industry level. Lastly, I compute the weighted average across industries, where I use total value added as the weight for each industry.²⁵

The second column of Table 4 show the average value of σ_R^2 between 2012 and 2016²⁶ for each country in the sample. The estimates range between 1.30 and 3.36, with a mean of 2.14, indicating a not negligible degree of dispersion.

Estimates for γ_{t+1} . The last step before estimating γ_{t+1} is assigning a value for ν . I set $\nu = 0.1$ to target a value of $\gamma_t = 0.1$ for Spain, which is at the lower end of the estimates of the effects of investment into TFP surveyed by González et al. (2024)²⁷ as it is not possible to separate how much of the reported effects are due to reduced misallocation as opposed to other channels such as innovation. The third column of Table 4 shows the estimated γ_{t+1} using this estimate for ν . A 1% increase in capital leads to increases in TFP between 0.07% and 0.13%.

Figure 1 shows the relationship between γ_{t+1} and the stock of capital and Gross National Income (GNI). As predicted by the formulation in Lemma 3, γ_{t+1} is decreasing in the stock of capital. Moreover, it is also decreasing in GNI, suggesting that it is inversely related to

²²Defined as firms that report NACE codes between 1010 and 3320

²³I use the same sample as Kalemli-Özcan et al. (2024) and drop the United Kingdom and Greece as some variables required for the analysis are not populated.

²⁴See Section C.2 in the Online Appendix.

²⁵Results are robust to using the unweighted measure.

²⁶I focus on the latter part of the sample to maximize coverage of manufacturing activity.

²⁷González et al. (2024) report that a 1 p.p drop in interest rates leads to an increase in TFP between 0.42 and 1.72. Assuming a semi-elasticity of investment of 4 (Wroblewski, 2025) implies $\gamma \in (0.1, 0.43)$

Table 4: Misallocation Estimates

Country	σ_R^2	γ_t
Austria	1.52	0.07
Belgium	1.54	0.07
Czech Republic	2.43	0.11
Estonia	1.93	0.08
Finland	1.58	0.07
France	1.33	0.06
Germany	1.58	0.07
Hungary	2.69	0.12
Italy	2.13	0.09
Latvia	2.86	0.13
Norway	1.99	0.09
Poland	2.35	0.10
Portugal	1.99	0.09
Romania	2.72	0.12
Slovak Republic	2.68	0.12
Slovenia	2.73	0.12
Spain	2.34	0.10
Sweden	2.32	0.10
Mean	2.15	0.09

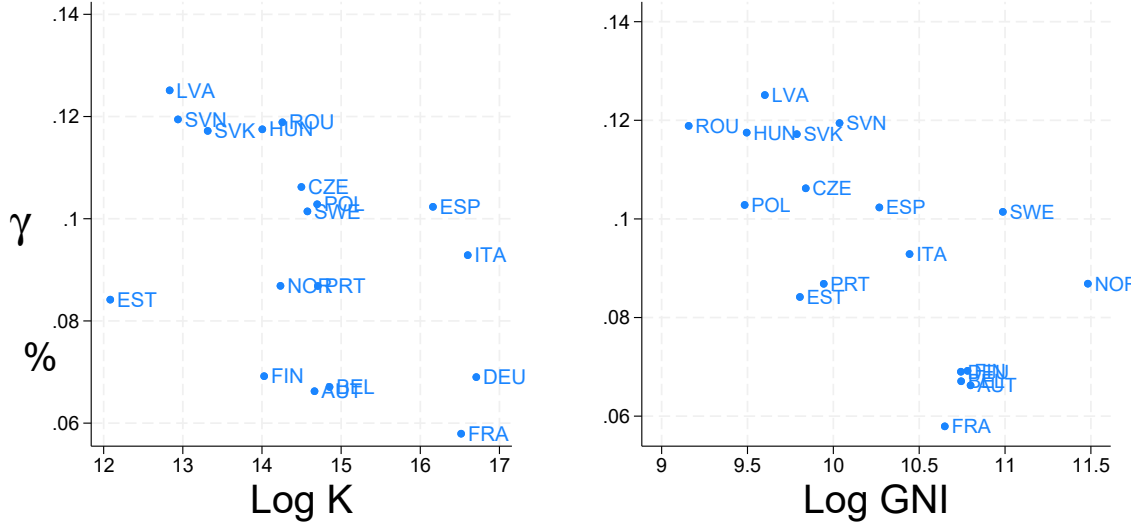
the degree of development of an economy.

To compute the second-best capital controls, I also consider an upper bound using the case of India. Attributing all the dispersion reported in Hsieh and Klenow (2009) in TFP to MRPK, that would result in $\sigma_R^2 = 5$, which in turn implies $\gamma_{t+1} = 0.52$.

Robustness As shown by equation (27), the estimation of γ depends on properly choosing ν and estimating σ_R^2 . In Online Appendix D.1.1, I estimate alternative values of ν using identified effects from Albrizio, Beatriz, and Khametsin (2024), Bau and Matray (2023) and Pinardon-Touati (2024) and show that they yield considerable higher estimates of ν . As such, I see the value of ν used in this paper as a conservative estimate.

In Online Appendix D.1.2, I show alternative estimates of γ using different ways of measuring σ_R^2 . I find that the main results are not driven by small firms, measured either by capital or number of workers, nor by young firms. Results are also robust to potential misreporting by firms, which I address by using averages within firm to estimate added value and the capital stock. Lastly, in Online Appendix D.2, I show that the estimates are also

Figure 1: Correlation of γ_{t+1} with GNI and the capital stock



Notes: The Figure plots the relationship between the estimates of γ_{t+1} , and the log of the capital stock (K) and Gross National Income (GNI), respectively, at the country-year level. The capital stock is obtained from Feenstra, Inklaar, and Timmer (2015) and is measured in PPP US dollars. GNI is sourced from World Bank (2021) and is measured in current US dollars using the Atlas method.

similar when allowing for labor misallocation.

6.3 Second-Best Capital Controls

It is now possible to quantify the second-best capital controls. Table 5 shows values of $\tau_t^{d, sb}$ for different combinations of γ_{t+1} and π . The former takes values that cover the empirical estimates discussed in the previous section, while the latter spans the values suggested by the literature.

The misallocation channel is quantitatively significant. Compared to the first row, a γ_{t+1} of 0.1, which is at the average of the estimates in the sample, implies a capital control that is negative and almost 0.30% lower. Capital controls are negative for crisis probabilities below 3%, suggesting that a borrowing subsidy is optimal during normal times. As the probability of a sudden stop increases, capital controls ramp up to values between 2.47% and 1.73% for the countries in the sample.

In addition to heterogeneity along π , there's also significant heterogeneity across levels of γ_{t+1} . The last row of table 5 shows that economies with a high level of misallocation find it optimal to subsidize borrowing even when the probability of a crisis is substantial. When considering the estimates for γ_{t+1} in Table 4, $\tau_t^{d, sb}$ ranges from 0.59% to -0.13% for $\pi = 6\%$.

Table 5: Estimates of $\tau_t^{d, sb}$

		π					
		1.7%	2.4%	3%	6%	14%	25%
$\gamma_{\tilde{t}}$	0	0.17	0.24	0.30	0.59	1.38	2.47
	0.05	-0.07	0.00	0.06	0.35	1.14	2.22
	.1	-0.31	-0.25	-0.19	0.11	0.89	1.97
	.15	-0.55	-0.49	-0.43	-0.13	0.65	1.73

Notes: The table shows values of the second-best capital control for different values of the crisis probability π and the productivity externality γ_{t+1} . The chosen range of values for π follows from the literature and is explained in Section 6. The range of values for γ_{t+1} is explained in Subsection 6.2.

What are the policy implications of these findings? Figure 2 plots values of $\tau_t^{d, sb}$ for different values of γ for $\pi = 0.03$ and compares it to the case where tradable production is an endowment. As a result, there are no negative effects on investment and productivity. For this probability of a crisis, the optimal capital control would be 0.3%. The solid blue line plots the $\tau_t^{d, sb}$ that takes into account the costs on both investment and productivity. To put these numbers into context, I show what the taxes would be for Germany, Spain and Latvia according to their estimated γ_{t+1} . In the case of Spain, which has an estimated γ_{t+1} close to the sample median, the tax on borrowing turns into a subsidy of almost 1%. Even for Germany, which has the lowest productivity cost out of the three, a subsidy of around -0.5% is optimal. In the case of Latvia, the subsidy would equal 1.2%, showing strong heterogeneity between the countries in the sample.

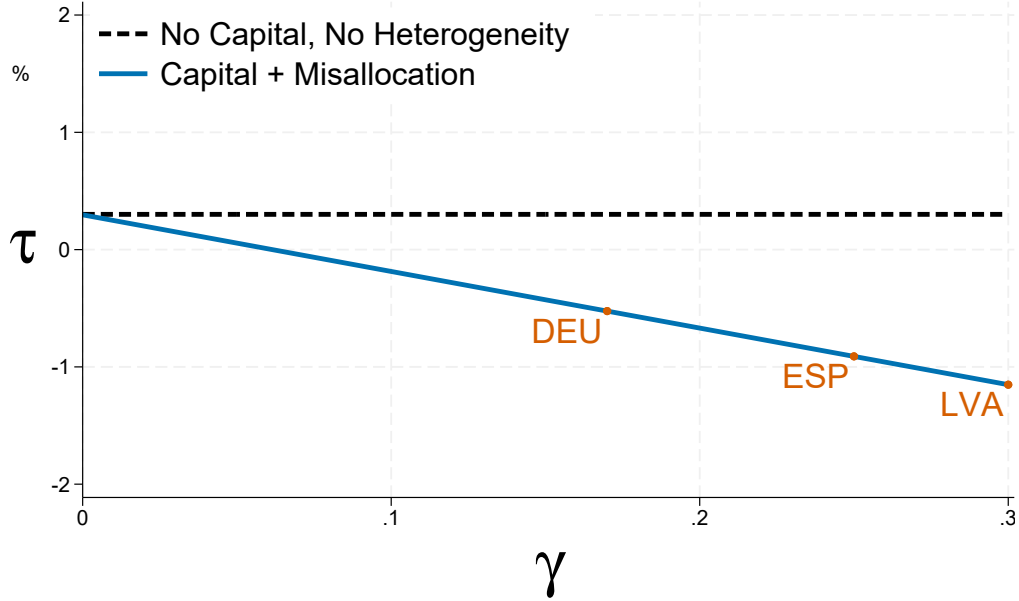
The results in this section show that the productivity losses inflicted by capital controls are quantitatively relevant, outweighing the benefit of macroprudential policies for a low probability of crises and plausible values for γ_{t+1} . This result hinges on the government not being able to control the allocation of credit, which renders macroprudential policy harmful by deterring capital accumulation.

7 Conclusion

In this paper, I study optimal macroprudential policy when its effects on productivity are considered. To accomplish this, I extend the standard model of macroprudential policy to incorporate production by firms that require physical capital and face heterogeneous frictions in the market for the latter.

I propose a tractable and flexible way of modeling these frictions, which result in capital being misallocated and, in turn, lower aggregate productivity as a result. Macroprudential

Figure 2: Comparison of First and Second-Best Taxes



Notes: The figure plots the optimal capital control under different scenarios for different values of γ and $\pi = 0.03$. The dashed black line plots $\tau_t^{d, sb}$ when tradable output is an endowment. The solid blue line shows $\tau_t^{d, sb}$ as a function of γ .

policies, such as capital controls, that reduce investment also exacerbate misallocation, in line with the empirical literature.

Turning to optimal policy, I first study the constrained efficient solution, in which the policymaker has enough available instruments. In this scenario, there's no trade-off, since the policymaker can tax borrowing to address the pecuniary externality and subsidize investment to improve productivity.

I then study the second-best problem, in which the policymaker can only control total borrowing. I characterize the solution to this problem and show that a trade-off exists. An increase in borrowing reduces welfare if a sudden stop occurs, but also leads to increased investment and productivity. I characterize this trade-off using several sufficient statistics, which allows me to provide intuition for the result and explain the qualitative effects of each of them.

To explore the quantitative relevance of this trade-off, I leverage the formulation of the second-best policy as a function of these sufficient statistics. Along with time-series moments and some calibrated moments from the literature, I exploit the tractable specification for misallocation to calibrate the relevant parameters using detailed firm-level data for a wide selection of European countries.

I find that capital effects have a sizable effect on productivity and that, as a result, the trade-off is quantitatively relevant. For the baseline probability of a crisis, potential productivity losses result in the optimal policy being a subsidy rather than a tax on capital inflows.

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A Proofs

A.1 Proof of Lemma 1

Proof. I prove this Lemma in two parts. First, I derive the aggregate production function and TFP. Lastly, I obtain the first order conditions of the representative firm.

Aggregate Production Function. Let MC be the marginal cost of the firm, derived from solving the cost minimization problem of the firm, with first order conditions:

$$R(i) = \alpha MC(i) \frac{y(i)}{k(i)} \quad (\text{A.1})$$

$$w = (1 - \alpha) MC(i) \frac{y(i)}{h(i)} \quad (\text{A.2})$$

Combining into the production function yields the marginal cost of the firm:

$$MC(i) = A(i)^{-1} \left(\frac{R(i)}{\alpha} \right)^\alpha \left(\frac{w}{1 - \alpha} \right)^{1-\alpha} \quad (\text{A.3})$$

We can now solve for the problem of the firm as:

$$\max_{y(i)} \frac{\eta}{\eta - 1} y(i)^{\frac{\eta-1}{\eta}} y^{\frac{1}{\eta}} - y(i) MC(i) \quad (\text{A.4})$$

with solution:

$$y(i) = (MC(i))^{-\eta} y \quad (\text{A.5})$$

Plugging back into the factor demands, we get:

$$k(i) = A(i)^{\eta-1} y \left(\frac{\alpha}{R(i)} \right)^{1+\alpha(\eta-1)} \left(\frac{1 - \alpha}{w} \right)^{(1-\alpha)(\eta-1)} \quad (\text{A.6})$$

$$h(i) = A(i)^{\eta-1} y \left(\frac{\alpha}{R(i)} \right)^{\alpha(\eta-1)} \left(\frac{1 - \alpha}{w} \right)^{\alpha+(1-\alpha)\eta} \quad (\text{A.7})$$

Then, we can obtain aggregate production as

$$y = \left[\int y(i)^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}} \quad (\text{A.8})$$

$$= \frac{\left[\int y(i)^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}}}{\left(\int k_t(i) di \right)^\alpha \left(\int h_t(i) di \right)^{1-\alpha}} k_t^\alpha h_t^{1-\alpha} \quad (\text{A.9})$$

$$= \frac{\left[\int A(i)^{\eta-1} R(i)^{-\alpha(\eta-1)} di \right]^{\frac{\eta}{\eta-1}}}{\left(\int (A(i))^{\eta-1} (R(i)^{-1})^{1+\alpha(\eta-1)} di \right)^\alpha \left(\int (A(i))^{\eta-1} (R(i)^{-1})^{\alpha(\eta-1)} di \right)^{1-\alpha}} k_t^\alpha h_t^{1-\alpha} \quad (\text{A.10})$$

Assuming either log-normality or up to second order, we can write:

$$y = TFP k^\alpha h^{1-\alpha} \quad (\text{A.11})$$

where²⁸

$$TFP = \left(\int A(i)^{\eta-1} \right)^{\frac{1}{\eta-1}} \exp \left[-\frac{1}{2} \alpha (1 + \alpha(\eta-1)) \sigma_R^2 \right] \quad (\text{A.12})$$

taking logs concludes the first part of the proof.

First order conditions. Aggregate labor and capital demand at the firm level,

$$k = y \alpha^{1+\alpha(\eta-1)} \left(\frac{1-\alpha}{w} \right)^{(1-\alpha)(\eta-1)} \int A(i)^{\eta-1} R(i)^{-(1+\alpha(\eta-1))} \quad (\text{A.13})$$

$$h = y \alpha^{\alpha(\eta-1)} \left(\frac{1-\alpha}{w} \right)^{\alpha+(1-\alpha)\eta} \int A(i)^{\eta-1} R(i)^{-\alpha(\eta-1)} \quad (\text{A.14})$$

divide by each other,

$$\frac{k}{h} = \alpha \frac{w}{1-\alpha} \exp \left(-\mathbb{E} [\log R(i)] + \frac{1}{2} (1 + 2\alpha(\eta-1)) \sigma_R^2 \right) \quad (\text{A.15})$$

plug back into (A.14),

$$1 = TFP^\eta \left(\frac{k}{h} \right)^{\alpha\eta} \left(\frac{1-\alpha}{w} \right)^\eta, \quad (\text{A.16})$$

solving for w ,

$$w = (1-\alpha) TFP \left(\frac{k}{h} \right)^\alpha \quad (\text{A.17})$$

In the same way, substitute $\frac{1-\alpha}{w}$ in (A.13)

$$\exp(\mathbb{E} [\log R(i)]) = \alpha \mathbb{E} [A(i)^{\eta-1}]^{\frac{1}{\eta-1}} \left(\frac{h}{k} \right)^{(1-\alpha)} \exp \left(\frac{1}{2} (1 - 3\alpha + \alpha^2 + 2\alpha\eta - \alpha^2\eta) \sigma_R^2 \right) \quad (\text{A.18})$$

Let $R \equiv \frac{\int R(i)k(i)}{\int k(i)}$,

$$R = \frac{\int A(i)^{\eta-1} y \alpha^{1+\alpha(\eta-1)} R(i)^{-\alpha(\eta-1)} \left(\frac{1-\alpha}{w} \right)^{(1-\alpha)(\eta-1)}}{\int A(i)^{\eta-1} y \left(\frac{\alpha}{R(i)} \right)^{1+\alpha(\eta-1)} \left(\frac{1-\alpha}{w} \right)^{(1-\alpha)(\eta-1)}} \quad (\text{A.19})$$

$$R = \exp \left(\mathbb{E} [\log R(i)] - \frac{1}{2} (1 + 2\alpha(\eta-1)) \sigma_R^2 \right) \quad (\text{A.20})$$

$$R = \frac{\exp(\mathbb{E} [\log R(i)])}{\exp \left(\frac{1}{2} (1 + 2\alpha(\eta-1)) \sigma_R^2 \right)} \quad (\text{A.21})$$

²⁸Note that the covariance between $A(i)$ and $R(i)$ drops out due to properties of the log-normal distribution between (A.10) and (A.12)

$$R = \frac{\alpha \mathbb{E} [A(i)^{\eta-1}]^{\frac{1}{\eta-1}} \left(\frac{h}{k}\right)^{(1-\alpha)} \exp\left(\frac{1}{2}(1-3\alpha+\alpha^2+2\alpha\eta-\alpha^2\eta)\sigma_R^2\right)}{\exp\left(\frac{1}{2}(1+2\alpha(\eta-1))\sigma_R^2\right)} \quad (\text{A.22})$$

$$R = \alpha \mathbb{E} [A(i)^{\eta-1}]^{\frac{1}{\eta-1}} \left(\frac{h}{k}\right)^{(1-\alpha)} \exp\left(\frac{1}{2}\alpha(1+\alpha(\eta-1))\sigma_R^2\right) \quad (\text{A.23})$$

$$R = \alpha TFP \left(\frac{h}{k}\right)^{(1-\alpha)} \quad (\text{A.24})$$

□

A.2 Proof of Lemma 2

Proof. To simplify the proof, I will solve the problem of a benevolent planner that must allocate capital to firms. This is analogous to the structure described in the main body because banks are assumed to be competitive. I solve this problem by backward induction. First, firms choose labor given capital.

$$\max_{h(i)} \left(A(i)k(i)^\alpha h(i)^{1-\alpha} \right)^{\frac{\eta-1}{\eta}} y^{\frac{1}{\eta}} - wh(i) \quad (\text{A.25})$$

with first order condition,

$$(1-\alpha) \frac{\eta-1}{\eta} (A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}} h(i)^{\frac{\alpha-\alpha\eta-1}{\eta}} = w \quad (\text{A.26})$$

$$h(i) = \left(\frac{(1-\alpha) \frac{\eta-1}{\eta} (A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}}}{w} \right)^{\frac{\eta}{1+\alpha(\eta-1)}} \quad (\text{A.27})$$

Which means the value of the firm $V(A(i), k(i))$, taking capital and productivity as given, is

$$\begin{aligned} V(A(i), k(i)) &= (A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}} y^{\frac{1}{\eta}} \left(\left((1-\alpha) \frac{\frac{\eta-1}{\eta} (A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}}}{w} \right)^{\frac{\eta}{1+\alpha(\eta-1)}} \right)^{(1-\alpha) \frac{\eta-1}{\eta}} \\ &\quad - w \left(\frac{(1-\alpha) \frac{\eta-1}{\eta} (A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}}}{w} \right)^{\frac{\eta}{1+\alpha(\eta-1)}} \\ &= \left((A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}} y^{\frac{1}{\eta}} \right)^{\frac{\eta}{1+\alpha(\eta-1)}} w^{-\frac{(1-\alpha)(\eta-1)}{1+\alpha(\eta-1)}} \left((1-\alpha) \frac{\eta-1}{\eta} \right)^{\frac{(1-\alpha)(\eta-1)}{1+\alpha(\eta-1)}} \\ &\quad - w^{-\frac{(1-\alpha)(\eta-1)}{1+\alpha(\eta-1)}} \left((1-\alpha) \frac{\eta-1}{\eta} (A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{1+\alpha(\eta-1)}} \end{aligned}$$

$$\begin{aligned}
&= \left((A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}} y^{\frac{1}{\eta}} \right)^{\frac{\eta}{1+\alpha(\eta-1)}} w^{-\frac{(1-\alpha)(\eta-1)}{1+\alpha(\eta-1)}} \\
&\quad \left(\left((1-\alpha)\frac{\eta-1}{\eta} \right)^{\frac{(1-\alpha)(\eta-1)}{1+\alpha(\eta-1)}} - \left((1-\alpha)\frac{\eta-1}{\eta} \right)^{\frac{\eta}{1+\alpha(\eta-1)}} \right) \\
&= \left((A(i)k(i)^\alpha)^{\frac{\eta-1}{\eta}} y^{\frac{1}{\eta}} \right)^{\frac{\eta}{1+\alpha(\eta-1)}} \left(\frac{(1-\alpha)\frac{\eta-1}{\eta}}{w} \right)^{\frac{(1-\alpha)(\eta-1)}{1+\alpha(\eta-1)}} \frac{1+\alpha(\eta-1)}{\eta} \\
V(A(i), k(i)) &= k(i)^{\frac{\alpha(\eta-1)}{1+\alpha(\eta-1)}} A(i)^{\frac{\eta-1}{1+\alpha(\eta-1)}} \Gamma,
\end{aligned}$$

where Γ collects all the terms that do not depend on $k(i)$ or $A(i)$.

Uncertainty Reveal. After choosing k and before choosing labor, $F(i)$ is revealed, such that

$$k(i) = \hat{k}(i) - A^{\nu(\eta-1)} F(i) \hat{k}(i)^{1-\nu} \quad (\text{A.28})$$

where $\hat{k}(i)$ is the capital chosen by the banks. Then, the value ex-ante V_e is given by

$$V_e(\hat{k}(i), A(i)) = \mathbb{E} \left[\left(\hat{k}(i) - A^{\nu(\eta-1)} F(i) \hat{k}(i)^{1-\nu} \right)^{\frac{\alpha(\eta-1)}{1+\alpha(\eta-1)}} A(i)^{\frac{\eta-1}{1+\alpha(\eta-1)}} \Gamma \right] \quad (\text{A.29})$$

Capital Choice. The allocation of capital is a solution to

$$\max_{k(i)} \int V_e(k(i), A(i)) \quad \text{s.t.} \quad \int k(i) = k \quad (\text{A.30})$$

with first order condition:

$$\begin{aligned}
&\hat{k}(i) \mathbb{E} \left[\left(1 - \frac{A^{\nu(\eta-1)} F(i)}{\hat{k}(i)^\nu} \right)^{\frac{1}{1+\alpha(\eta-1)}} \left(1 - (1-\nu) \frac{A^{\nu(\eta-1)} F(i)}{\hat{k}(i)^\nu} \right) \right]^{-(1+\alpha(\eta-1))} \\
&= A(i)^{\eta-1} \left(\frac{1+\alpha(\eta-1)}{\alpha(\eta-1)} \Gamma \lambda \right)^{-(1+\alpha(\eta-1))}
\end{aligned} \quad (\text{A.31})$$

Dividing by the same condition for firm j ,

$$\begin{aligned} & \frac{\hat{k}(i)}{\hat{k}(j)} \left(\frac{\mathbb{E} \left[\left(1 - \frac{A(i)^{\nu(\eta-1)} F(i)}{\hat{k}(i)^\nu} \right)^{\frac{1}{1+\alpha(\eta-1)}} \left(1 - (1-\nu) \frac{A(i)^{\nu(\eta-1)} F(i)}{\hat{k}(i)^\nu} \right) \right]^{-(1+\alpha(\eta-1))}}{\mathbb{E} \left[\left(1 - \frac{A(j)^{\nu(\eta-1)} F(j)}{\hat{k}(j)^\nu} \right)^{\frac{1}{1+\alpha(\eta-1)}} \left(1 - (1-\nu) \frac{A(j)^{\nu(\eta-1)} F(j)}{\hat{k}(j)^\nu} \right) \right]^{-(1+\alpha(\eta-1))}} \right)^{-(1+\alpha(\eta-1))} \\ &= \left(\frac{A(i)}{A(j)} \right)^{\eta-1} \end{aligned} \tag{A.32}$$

I guess and later verify that $k(i) = \bar{A} A(i)^{\eta-1} k$, where $\bar{A}$ is a constant.

$$\begin{aligned} & \left(\frac{A(i)}{A(j)} \right)^{\eta-1} \left(\frac{\mathbb{E} \left[\left(1 - \frac{F(i)}{(Ak)^\nu} \right)^{\frac{1}{1+\alpha(\eta-1)}} \left(1 - (1-\nu) \frac{F(i)}{(Ak)^\nu} \right) \right]^{-(1+\alpha(\eta-1))}}{\mathbb{E} \left[\left(1 - \frac{F(j)}{(Ak)^\nu} \right)^{\frac{1}{1+\alpha(\eta-1)}} \left(1 - (1-\nu) \frac{F(j)}{(Ak)^\nu} \right) \right]^{-(1+\alpha(\eta-1))}} \right)^{-(1+\alpha(\eta-1))} \\ &= \left(\frac{A(i)}{A(j)} \right)^{\eta-1} \end{aligned} \tag{A.33}$$

$$1 = 1 \tag{A.34}$$

To find $\bar{A}$, plug into the resource constraint,

$$\int k(i) = k \tag{A.35}$$

$$\int \bar{A} A(i)^{\eta-1} k = k \tag{A.36}$$

$$\bar{A} k \int A(i)^{\eta-1} = k \tag{A.37}$$

$$\bar{A} = \mathbb{E} [A(i)^{\eta-1}]^{-1} \tag{A.38}$$

□

A.3 Proof of Lemma 3

Proof. Start from capital demand (A.6) and solve for $R(i)$ as a function of $k(i)$,

$$R(i) = \alpha \left(k(i)^{-1} A(i)^{\eta-1} y \left(\frac{1-\alpha}{w} \right)^{(1-\alpha)(\eta-1)} \right)^{\frac{1}{1+\alpha(\eta-1)}} \tag{A.39}$$

Combining with Lemma 2,

$$R(i) = \alpha \left(k^{-1} \left(1 + \frac{F(i)}{k^\nu} \right)^{-1} y \left(\frac{1-\alpha}{w} \right)^{(1-\alpha)(\eta-1)} \right)^{\frac{1}{1+\alpha(\eta-1)}}, \quad (\text{A.40})$$

taking logs,

$$\log R(i) = \log \alpha + \frac{1}{1+\alpha(\eta-1)} \left(-\log k - \log \left(1 + \frac{F(i)}{k^\nu} \right) + \log y + \log \left(\frac{1-\alpha}{w} \right)^{(1-\alpha)(\eta-1)} \right), \quad (\text{A.41})$$

Lastly, using the assumption of small $F(i)$,

$$\log R(i) = \log \alpha + \frac{1}{1+\alpha(\eta-1)} \left(-\log k - \frac{F(i)}{k^\nu} + \log y + \log \left(\frac{1-\alpha}{w} \right)^{(1-\alpha)(\eta-1)} \right). \quad (\text{A.42})$$

It follows that

$$\mathbb{V}ar [\log R(i)] = \left(\frac{1}{1+\alpha(\eta-1)} \frac{1}{k^\nu} \right)^2 \sigma_F^2 \quad (\text{A.43})$$

□

A.4 Proof of Proposition 1

Proof. I need to show that conditions (4),(30),(31),(34) and (35) hold. The constraints holds trivially. Set τ^d so that (4) holds,

$$(1 + \tau_t^d) \mathbb{E} \left[\frac{c_{T,t+1}^{-\sigma}}{p_{t+1}} \right] = \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \left(1 + \kappa \mu_{t+1} \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right) \right] \quad (\text{A.44})$$

$$\tau_t^d \mathbb{E} \left[\frac{c_{T,t+1}^{-\sigma}}{p_{t+1}} \right] = \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \kappa \mu_{t+1} \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right] \quad (\text{A.45})$$

$$\tau^d = \mathbb{E} \left[\frac{c_{T,t+1}^{-\sigma}}{p_{t+1}} \right]^{-1} \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \kappa \mu_{t+1} \times \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right] \quad (\text{A.46})$$

and τ^k so that (35) holds,

$$\begin{aligned} & \mathbb{E} \left[c_{t+1}^{-\sigma} \left(\frac{\alpha \text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{p_{t+1}} (1 + \kappa \mu_{t+1}) + \frac{\tau_{t+1}^k}{p_{t+1}} \right) \right] \\ &= \mathbb{E} \left[c_{t+1}^{-\sigma} \left((\alpha + \gamma_{t+1}) \frac{\text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{p_{t+1}} \right) \left(1 + \kappa \mu_{t+1} \left(1 + \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right) \right) \right] \end{aligned} \quad (\text{A.47})$$

$$\begin{aligned} & \tau_{t+1}^k \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right] \\ &= \mathbb{E} \left[c_{t+1}^{-\sigma} \left(\frac{\text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{p_{t+1}} \right) \left(\gamma_{t+1}(1 + \kappa \mu_{t+1}) + (\alpha + \gamma_{t+1}) \kappa \mu_{t+1} \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right) \right] \end{aligned} \quad (\text{A.48})$$

$$\begin{aligned} \tau^k &= \\ & \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right]^{-1} \mathbb{E} \left[c_{t+1}^{-\sigma} \left(\frac{\text{TFP}_{t+1} k_{t+1}^{\alpha-1}}{p_{t+1}} \right) \left(\gamma_{t+1}(1 + \kappa \mu_{t+1}) + (\alpha + \gamma_{t+1}) \kappa \mu_{t+1} \frac{1}{\xi} \frac{p_{t+1}^N y^N}{x_{t+1}^T} \right) \right] \end{aligned} \quad (\text{A.49})$$

□

A.5 Proof of Proposition 2

Proof. Putting together (4) and (42)

$$\begin{aligned} (1+r)(1+\tau_t^d) \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right] &= \beta \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} (1+r \right. \\ & \quad \left. + \kappa \mu_{t+1} ((1+r) - \text{mpi}((\alpha + \gamma_{t+1}) \text{TFP}_{t+1} k_{t+1}^{\alpha-1})) \frac{1}{\xi} \frac{p_{t+1} y^N}{x_{t+1}^T} \right. \\ & \quad \left. - \text{mpi} \gamma_{t+1} \text{TFP}_{t+1} k_{t+1}^{\alpha-1} (1 + \kappa \mu_{t+1})) \right] \end{aligned} \quad (\text{A.50})$$

$$\begin{aligned} \tau_t^d \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \right] &= \beta \mathbb{E} \left[\frac{c_{t+1}^{-\sigma}}{p_{t+1}} \left(\kappa \mu_{t+1} \left(1 - \frac{\text{mpi}}{1+r} ((\alpha + \gamma_{t+1}) \text{TFP}_{t+1} k_{t+1}^{\alpha-1}) \right) \frac{1}{\xi} \frac{p_{t+1} y^N}{x_{t+1}^T} \right. \right. \\ & \quad \left. \left. - \frac{\text{mpi}}{1+r} \gamma_{t+1} \text{TFP}_{t+1} k_{t+1}^{\alpha-1} (1 + \kappa \mu_{t+1}) \right) \right] \end{aligned} \quad (\text{A.51})$$

Using the definitions (39) and (40) completes the proof. □

B Capital Misallocation Block

In this section I explore some alternative microfoundations and show how to generalize the model to accommodate additional sources of misallocation.

B.1 Possible Microfoundations

I know explore three possible microfoundations that span three commonly mentioned sources of misallocation: Credit shocks, borrowing constraints, and equity constraints.

B.1.1 Credit shocks

In this model, the frictions $F(i)$ represent shocks to the credit supply of firms, as in the models of Chodorow-Reich (2013), Herreño (2023), and Pinardon-Touati (2024). These shocks can be overhead costs of the bank, monitoring or operating costs, additional sources of demand that reduce available credit to firms such as government credit demand or changes in the balance sheet of banks.

No matter the source, I assume that these shocks $S(i)$ take the form

$$S(i) = A^{\eta-1} F(i) k^\gamma \quad (\text{B.1})$$

Where $F(i)$ is a random variable that represents different realizations in these shocks across firms. For instance, a firm that is more geographically remote or more opaque will imply higher costs for the bank. Likewise, different firms might be served by banks with different balance sheets, or they might be located in regions with different government spending. What is important is that the magnitude of these shocks do not scale perfectly with the total capital available to the bank. This is achieved by $0 < \gamma < 1$.

Let $\nu = 1 - \gamma$, then, as a result, the amount banks can lend is given by

$$k(i) = \frac{A(i)^{\eta-1}}{\mathbb{E}[A(i)]^{\eta-1}} k \left(1 - \frac{F(i)}{\mathbb{E}[A(i)^{\eta-1}]^\nu k^\nu} \right) \quad (\text{B.2})$$

B.1.2 Borrowing Constraints

In this model, I assume that the household in charge of the bank can divert a fraction of the credit. If they do that, the other households can seize a fraction of their capital k . I follow the same timing as Bianchi and Mendoza (2018) and assume that this diversion happens before any other trading occurs. Once the capital is seized, I assume that the household buys it again.

After diverting the credit, the household can hide a fraction $A(i)^{\nu(\eta-1)} F(i) k(i)^{-\nu}$ of its assets, where $F(i)$ is the idiosyncratic component. While in this case it is a characteristic of the household, it can also be interpreted as describing the firm level of opaqueness or perceived riskiness.

It follows that, to avoid any diversion in equilibrium, the following incentive compatibility must hold,

$$k(i) \leq \frac{A(i)^{\eta-1}}{\mathbb{E}[A(i)]^{\eta-1}} k \left(1 - \frac{F(i)}{\mathbb{E}[A(i)^{\eta-1}]^\nu k^\nu} \right) \quad (\text{B.3})$$

In equilibrium, (B.3) holds with equality.

B.1.3 Internal and External Finance

In this setup, firms assemble capital by combining bank credit and equity $E(i)$ in the following way,

$$k(i) = \left(\theta k_b(i)^{\frac{\rho-1}{\rho}} + (1-\theta)(A(i)^{\eta-1}E(i))^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (\text{B.4})$$

The CES specification is a reduced form way of modelling the imperfect substitutability between both sources of financing, given by ρ in this case.

For simplicity, I assume that Equity is fixed, although it would be possible to have it be a choice of the firm. Then, total capital can be written as

$$k(i) = \theta k_b(i) \left(1 + \frac{1-\theta}{\theta} \left(\frac{A(i)^{\eta-1}E(i)}{k_b(i)} \right)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (\text{B.5})$$

Using Lemma 2,

$$k(i) = \theta \frac{A(i)^{\eta-1}}{\mathbb{E}[A(i)^{\eta-1}]} k \left(1 + \frac{1-\theta}{\theta} \left(\frac{E(i)}{\frac{k}{\mathbb{E}[A(i)^{\eta-1}]}} \right)^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (\text{B.6})$$

$$(\text{B.7})$$

Let $F(i) \equiv \left(\frac{1-\theta}{\theta} \mathbb{E}[A(i)^{\eta-1}] E(i) \right)^{\frac{\rho-1}{\rho}}$ and $\nu \equiv \frac{\rho-1}{\rho}$ to write

$$k(i) = \theta \frac{A(i)^{\eta-1}}{\mathbb{E}[A(i)^{\eta-1}]} k \left(1 + \frac{F(i)}{k^\nu} \right)^{\frac{1}{\nu}}, \quad (\text{B.8})$$

which yields the desired result.

B.2 Including Additional Sources of Misallocation

In this section, I show that the model can accommodate other sources of misallocation that are not related to investment. To keep things simple, I consider only one additional distortion: Firms face a labor wedge τ^w .

Taking the stock of capital $k(i)$ as given, the problem of an individual firm is given by

$$\max_{h(i)} y^{\frac{1}{\eta}} \left(A(i)k(i)^\alpha h(i)^{1-\alpha} \right)^{\frac{\eta-1}{\eta}} - (1 + \tau^w(i))wh(i), \quad (\text{B.9})$$

All other elements of the model remain the same, including the timing: Banks select the supply to the market populated by firm i before knowing the shocks $F(i)$ and the realization

of the labor wedge $\tau^w(i)$.

As a result, and following a logic similar to Lemma 2, the firm's cost of capital is described by

$$R(i) \propto \left(\left(1 - \frac{F(i)}{k^\nu} \right) ((1 + \tau^w(i))w)^{(1-\alpha)(\eta-1)} \right)^{\frac{1}{1+\alpha(\eta-1)}} \quad (\text{B.10})$$

Let $\sigma_{\tau^w}^2$ be the variance of $\tau^w(i)$ and σ_{F,τ^w} its covariance with $F(i)$. Then, the variance of $\log R(i)$ is given by

$$\sigma_R^2 = \frac{1}{(1 + \alpha(\eta - 1))^2} \left(\frac{\sigma_F^2}{k^{2\nu}} + ((1 - \alpha)(\eta - 1))^2 \sigma_{\tau^w}^2 + 2 \frac{(1 - \alpha)(\eta - 1)}{k^\nu} \sigma_{F,\tau^w} \right) \quad (\text{B.11})$$

It follows from (B.11) that the elasticity of TFP with respect to capital is now given by

$$\gamma = \nu \frac{\alpha}{1 + \alpha(\eta - 1)} \left(\frac{\sigma_F^2}{k^{2\nu}} + \frac{(1 - \alpha)(\eta - 1)}{k^\nu} \sigma_{F,\tau^w} \right) \frac{1}{k} \quad (\text{B.12})$$

Note that now there's not a direct link to an empirical moment. Identification, however, is made possible by considering two additional moments: Dispersion in the marginal revenue productivity of labor (mrpl) and the capital to labor ratio, with the former defined as

$$\text{mrpl} \propto \frac{(A(i)k(i)^\alpha h(i)^{1-\alpha})^{\frac{\eta-1}{\eta}}}{h(i)} \quad (\text{B.13})$$

Solving for optimal labor in (B.9), it is easy to see that

$$\log \text{mrpl} \propto \log 1 + \tau^w(i) \quad (\text{B.14})$$

And so it follows that the variance of the $\log \text{mrpl}$ identifies $\sigma_{\tau^w}^2$,

$$\text{Var} [\log \text{mrpl}] = \sigma_{\tau^w}^2 \quad (\text{B.15})$$

Similarly, the variance of the $\log$ capital to labor ratio is given by

$$\text{Var} \left[\log \frac{k(i)}{h(i)} \right] = \frac{1}{(1 + \alpha(\eta - 1))^2} \left(\eta^2 \sigma_{\tau^w}^2 + \frac{F(i)}{k^{2\nu}} - 2 \frac{\eta}{k^\nu} \sigma_{F,\tau^w} \right) \quad (\text{B.16})$$

Then, after using (B.15), (B.16) and (B.11) define a system of two equations and two un-

knowns. Solving for $\frac{\sigma_{F,\tau^w}}{k^\nu}$ yields

$$\frac{\sigma_{F,\tau^w}}{k^\nu} = \frac{(1 + \alpha(\eta - 1))^2(\mathbb{V}ar[mrpk] - \mathbb{V}ar\left[\log \frac{k(i)}{h(i)}\right]) + \sigma_{\tau^w}^2(\eta^2 - ((1 - \alpha)(\eta - 1))^2)}{2(\eta + (1 - \alpha)(\eta - 1))} \quad (\text{B.17})$$

and $\frac{\sigma_F^2}{k^{2\nu}}$ is given by

$$\frac{\sigma_F^2}{k^{2\nu}} = (1 + \alpha(\eta - 1))^2 \mathbb{V}ar\left[\log \frac{k(i)}{h(i)}\right] - \eta^2 \sigma_{\tau^2}^2 + 2\frac{\eta}{k^\nu} \sigma_{F,\tau^w} \quad (\text{B.18})$$

Together, they pin down γ in (B.12).

C Data Appendix

In this section I provide more details into the data work on Section 6.

C.1 Country Panel Data

Table C.1 shows summary stats for the sample of countries used to calculate the time series moments. The sample has 19 middle-income and 17 high-income economies.

C.2 Orbis

I access the Orbis dataset through the Orbis Project at the CBS grid, where it was made available through the efforts of Kochen (2022).

Table C.2 presents summary statistics for the 18 countries selected in the sample, which follow the sample from Kalemli-Özcan et al. (2024). Table C.3 shows the distribution of output and employment across the firm-size distribution. The results are similar to those reported by Kalemli-Özcan et al. (2024)²⁹, who compare it to aggregate statistics from Eurostat and find that Orbis provided a representative sample.

²⁹Table D.2.5

Table C.1: Summary Statistics for Panel of Countries

Country	First Year	Last Year	σ_Y	σ_C/σ_Y	$\sigma_{CA/Y}$	Sudden Stops
<i>Middle-Income</i>						
Argentina	1979	2012	9.75	1.19	3.05	3
Brazil	1979	2012	9.73	1.38	2.01	1
Chile	1979	2012	11.43	1.18	3.37	2
Colombia	1979	2012	5.74	1.48	2.84	1
Czech Republic	1979	2012	5.64	0.95	1.78	1
Dominican Republic	1979	2012	7.40	0.86	3.11	1
Ecuador	1979	2012	7.58	1.37	3.00	2
Egypt	1979	2012	3.98	0.75	3.35	0
El Salvador	1979	2012	8.17	1.64	2.45	1
Hungary	1979	2012	13.95	0.57	3.35	0
Indonesia	1979	2012	7.01	0.81	2.38	1
Malaysia	1979	2012	5.63	1.53	7.59	1
Mexico	1979	2012	6.44	1.14	2.32	2
Morocco	1979	2012	4.19	1.12	2.85	4
Pakistan	1979	2012	4.24	0.85	2.49	1
Peru	1979	2012	13.75	0.84	2.98	2
Philippines	1979	2012	10.13	0.48	2.74	1
Portugal	1979	2012	4.84	1.49	4.86	1
South Africa	1979	2012	9.56	0.91	2.67	2
South Korea	1979	2012	7.18	1.38	3.49	1
Tunisia	1979	2012	6.23	2.10	2.38	1
Ukraine	1979	2012	25.39	1.13	4.88	0
Venezuela	1979	2012	8.96	1.83	6.64	1
<i>High-Income</i>						
Canada	1979	2012	3.99	0.91	2.10	1
Denmark	1979	2012	3.04	0.97	1.54	1
Finland	1979	2012	6.45	0.82	3.44	0
France	1979	2012	2.44	1.09	1.05	1
Germany	1979	2012	2.42	1.00	2.26	0
Iceland	1979	2012	7.32	1.18	5.99	2
Italy	1979	2012	2.58	1.01	1.72	1
Netherlands	1979	2012	4.34	1.39	1.60	1
Norway	1979	2012	3.60	1.36	4.04	2
Spain	1979	2012	5.42	1.22	2.84	0
Sweden	1979	2012	4.48	1.02	2.34	1
Switzerland	1979	2012	2.54	0.58	2.44	0
United Kingdom	1979	2012	5.07	1.17	1.35	1
United States	1979	2012	3.01	1.10	1.30	2

Notes: This table presents summary statistics for the sample of countries used to estimate time-series moments in Section 6. σ_Y , σ_C and $\sigma_{CA/Y}$ are the standard deviations of log GDP per capita, log consumption per capita and the current account over GDP. All three variables are quadratically detrended. Sudden stops refer to the number of Sudden Stops as identified by Korinek and Mendoza (2014).

Table C.2: Summary Statistics for Countries in Orbis Sample

Country	N. Firms	N. Obs	Total Employees
Austria	9,283	88,004	72,401
Belgium	23,074	255,545	424,162
Czech Republic	42,064	314,829	823,799
Estonia	7,278	61,552	60,523
Finland	13,570	122,259	199,776
France	133,919	1,312,707	1,449,405
Germany	69,432	622,434	1,711,145
Hungary	144,485	1,084,401	483,002
Italy	204,584	1,659,439	2,042,621
Latvia	9,467	71,653	92,640
Norway	16,938	116,000	45,704
Poland	29,400	199,855	776,776
Portugal	55,038	462,410	352,406
Romania	62,621	539,698	807,445
Slovak Republic	22,344	136,706	190,646
Slovenia	20,231	138,929	116,640
Spain	158,426	1,634,164	1,520,580
Sweden	28,456	322,773	389,409

Notes: The first column shows the number of unique firms per country in the data. The second column and third column show the average number across years of observations and total employees respectively.

Table C.3: Size Distribution in the Manufacturing Sector: Total Sample, 2006

Panel A: Gross Output																		
	AT	BE	CZ	DE	EE	ES	FI	FR	HU	IT	LV	NO	PL	PT	RO	SE	SI	SK
1 to 19 employees	0.05	0.03	0.07	0.01	0.14	0.20	0.07	0.09	0.04	0.13	0.03	0.16	0.02	0.20	0.13	0.21	0.12	0.16
20 to 249 employees	0.38	0.40	0.45	0.24	0.61	0.60	0.41	0.45	0.62	0.60	0.51	0.51	0.33	0.63	0.48	0.42	0.44	0.45
250+ employees	0.57	0.57	0.49	0.75	0.25	0.20	0.52	0.46	0.34	0.27	0.46	0.32	0.66	0.17	0.39	0.37	0.45	0.39
Panel B: Employment																		
1 to 19 employees	0.16	0.11	0.05	0.01	0.12	0.23	0.09	0.09	0.03	0.11	0.13	0.23	0.01	0.24	0.11	0.16	0.09	0.08
20 to 249 employees	0.38	0.40	0.39	0.30	0.56	0.47	0.41	0.33	0.31	0.50	0.54	0.48	0.34	0.53	0.35	0.33	0.39	0.38
250+ employees	0.45	0.50	0.55	0.69	0.32	0.30	0.49	0.58	0.65	0.38	0.33	0.29	0.65	0.22	0.54	0.51	0.52	0.53

Notes: This table presents the firm size distribution of output and employment for the countries in the sample. I use total operating revenue as the definition of revenue at the firm level. The year 2006 is chosen for comparability to Kalemli-Özcan et al. (2024). The countries are Austria (AT), Belgium (BE), Czech Republic (CZ), Germany (DE), Estonia (EE), Spain (ES), Finland (FI), France (FR), Hungary (HU), Italy (IT), Latvia (LV), Norway (NO), Poland (PO), Portugal (PT), Romania (RO), Sweden (SE), Slovenia (SI) and (Slovakia).

D Robustness

D.1 TFP Elasticity with respect to Investment

The elasticity of TFP with respect to investment, γ , is a key object for the results of the paper, as it drives the costs of macroprudential policy in the quantitative exercise. In this section, I explore a few different ways of assigning a value to this parameter.

Recall from (27) that γ is given by

$$\alpha(1 + \alpha(\eta - 1))\nu\sigma_R^2$$

I fix the values of α and η following the literature and estimate σ_R^2 in the data, while I set $\nu = 0.1$ by targeting $\gamma = 0.1$ for Spain. In this section I perform robustness checks for the latter two.

D.1.1 Frictions Elasticity ν

Table D.1 presents estimates for ν using empirical findings from 4 different papers. My baseline estimate, $\nu = 0.1$, is significantly smaller than any of the estimates that find a positive relationship between investment and productivity growth. When using the results

Table D.1: Summary of ν Estimates

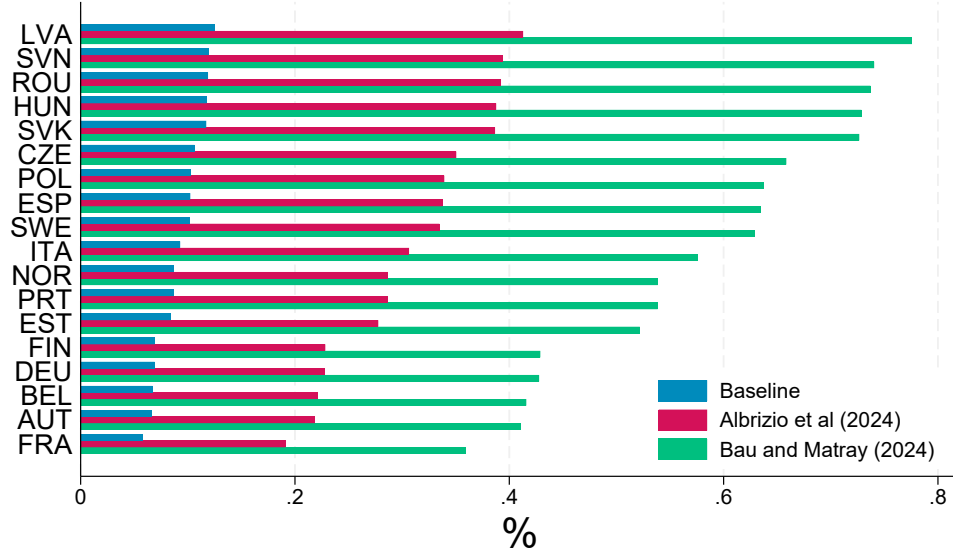
Estimate	Source
0.22	Pinardon-Touati (2024)
0.33	Albrizio, Beatriz, and Khametsin (2024)
0.62	Bau and Matray (2023)
-0.11	Gopinath et al. (2017)

Notes: This table summarizes the different estimates of the parameter ν considered in the analysis. See Section D.1.1 for details on their calculation.

from Gopinath et al. (2017), the estimates flip signs. It is important to keep in mind, that their work captures misallocation dynamics during a credit boom while this paper focuses on the effects of policy changes.

Figure D.1 plots the comparison between my baseline γ estimates (blue) to those obtained using the estimates of Albrizio, Beatriz, and Khametsin (2024) (red) and Bau and Matray (2023) (green).

Figure D.1: γ estimates for different ν .



Notes: The figure plots estimates for γ using the estimates from ν in Table D.1. Section 6 explains how to obtain estimates for γ , while Section D.1.1 details the source of the alternative estimates.

Albrizio, Beatriz, and Khametsin (2024) Using Spanish data with similar coverage to Orbis³⁰, Albrizio, Beatriz, and Khametsin (2024) study the causal effects of monetary policy shocks on misallocation. In their baseline results, a one standard deviation monetary shock leads to an increase in the capital stock of the average firm of around 1.2%. At the same, they find a reduction in the variance of log MRPK (σ_R^2) of 0.8%. Using Lemma 3 and taking logs, the result can be written as

$$\Delta \log \sigma_R^2 = -2 \times \nu \times \Delta \log k \quad (\text{D.1})$$

Using their results, it follows that $\nu = 0.33$.

Bau and Matray (2023) This article exploits a set of reforms in India that made it easier for foreign capital to access firms. To identify their effects, they leverage the staggered liberalization across industries and also the difference between firms with high and low marginal revenue productivity of capital (*mrpk*).

As shown in A.3, it is possible to write the *mrpk* of firm i as

³⁰Central de Balances Integrada and Directorio Central de Estadística

$$\log \text{mrpk}(i) \propto -\frac{1}{1 + \alpha(\eta - 1)} \left(\log k - \frac{F(i)}{k^\nu} \right) \quad (\text{D.2})$$

To obtain the relative change between firms, take the difference for firms i and j ,

$$\log \text{mrpk}(i) - \log \text{mrpk}(j) = \frac{1}{1 + \alpha(\eta - 1)} \frac{F(i) - F(j)}{k^\nu} \quad (\text{D.3})$$

and take the derivative with respect to $\log k$,

$$\frac{\partial \log \text{mrpk}(i)}{\partial \log k} - \frac{\partial \log \text{mrpk}(j)}{\partial \log k} = \frac{\nu}{1 + \alpha(\eta - 1)} \left(\frac{F(j) - F(i)}{k^\nu} \right) \quad (\text{D.4})$$

$$\frac{\partial \log \text{mrpk}(i)}{\partial \log k} - \frac{\partial \log \text{mrpk}(j)}{\partial \log k} = -\nu (\log \text{mrpk}(i) - \log \text{mrpk}(j)) \quad (\text{D.5})$$

which links the relative changes in mrpk to changes in total capital. Then,

$$\Delta \log \text{mrpk}(i) - \Delta \log \text{mrpk}(j) = -\nu (\log \text{mrpk}(i) - \log \text{mrpk}(j)) \times \Delta \log k \quad (\text{D.6})$$

Bau and Matray (2023) report³¹ a 32% increase in capital and that firms with high mrpk reduced their mrpk by 32% more than low mrpk firms. They also mention that the former originally had a mrpk 160% higher than the latter. It follows that $\nu = 0.62$. Using $\nu = 0.22$, equation (D.6) predicts an 11% decrease in the mrpk of high mrpk firms relative to low mrpk firms³². Of note is also the fact that they identify the effect of foreign capital entering into the market, which potentially improves the ability of the financial system. In the model, this would translate into an increase in ν , which here I take as a primitive.

Pinardon-Touati (2024) Studying the effect of government spending on private borrowing in France, she finds that a 0.28% decrease in capital leads on average to a 0.04% decrease in TFP, which corresponds to $\gamma = 0.14$. Given a $\sigma_R^2 = 1.32$ for France, that yields $\nu = 0.22$.

Gopinath et al. (2017) In their seminal paper, Gopinath et al. (2017) documented an increase in mrpk dispersion occurring at the same time as declines in real interest rate declines in several Southern European economies.

In the first column of Table V, they report the change in the standard deviation of mrpk

³¹See Table 4 (Bau and Matray, 2023).

³²This value is slightly larger than the upper endpoint of the 95% confidence interval for their estimate.

and the total change in the capital stock. Given Lemma 3,

$$\Delta\sigma_R = -\nu \times \sigma_R \times \Delta \log k$$

Gopinath et al. (2017) report $\Delta\sigma_R = 0.034$ and $\Delta \log k = 0.22$. While they do not report the initial value of σ_R , I can use the value reported by Albrizio, Beatriz, and Khametsin (2024), who study the same period, and is approximately 1.6. As a result, $\nu = -0.1$.

D.1.2 Capital Misallocation σ_R^2

In this section, I perform robustness checks over the estimation of σ_R^2 , which is one of the key inputs for estimating γ . First, I consider more restricted samples to ensure that no group particularly drives the results. Second, I use within-firm averages to mitigate potential measurement error.

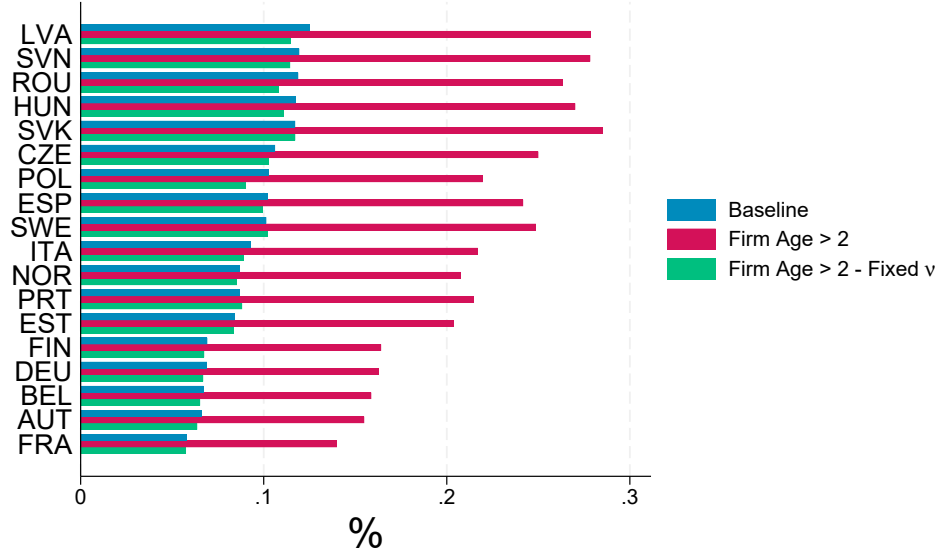
Figure D.2 plots estimates of γ when considering only firms older than 2 years to estimate capital misallocation. In blue I show the estimates when modifying ν to maintain $\gamma = 0.1$ for Spain. For comparison, I also plot in green the estimates while keeping ν unchanged. Even when keeping ν fixed the results do not change significantly, showing that results are not driven by young firms having high dispersion in returns to capital. Figure D.3 plots estimates of γ for the sample of firms with more than 5 employees. While there's a modest reduction in the estimates for countries in the upper end of the estimates, the overall message remains that these costs are significant and that results are not driven by small firms. Figure D.4 plots estimates of γ when excluding firms which report a stock of capital larger than 10,000 euros, as a way of ensuring that results are not driven by very small firms. Although the estimates change, the picture remains virtually the same when it comes to the magnitude of the results.

To mitigate potential measurement from firms misreporting data, I estimate σ_R^2 by using a within-firm moving average of their added value and capital stock. Figure D.5 plots the implied γ by country (red) compared to the baseline estimates (blue). Figure D.5 shows that applying this procedure modestly shrinks the estimates. For the median country, the resulting γ is 12% smaller than the baseline estimate. The main result that borrowing subsidies are optimal when the probability of a crisis is low still apply for these estimates.

D.2 Model With Additional Distortions

In this section I estimate γ according to the model in Section B.2. Following the identification argument in that section, I estimate $\text{Var}[\log \text{mrpl}(i)]$ and $\text{Var}\left[\log \frac{k(i)}{h(i)}\right]$ similarly to the

Figure D.2: γ Estimates for Firms older than 2 years



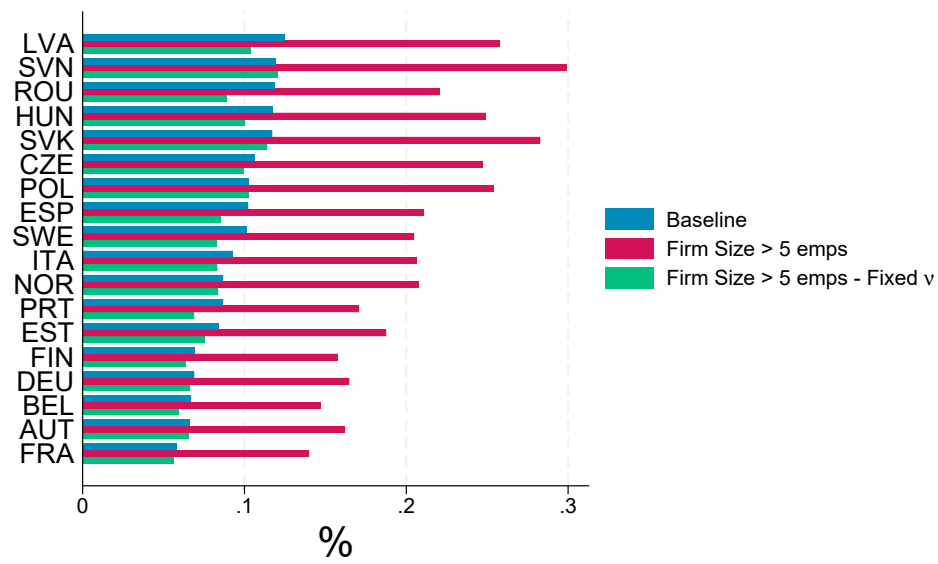
Notes: The figure plots the elasticity of TFP with respect to investment γ for a sample of countries according to (27). The blue bars plot the baseline estimates, explained in Section 6. The red bars plot the estimates using only firms older than 2 years to calculate capital misallocation. The green bar plots the estimates using only firms older than 2 years to calculate capital misallocation and keeping ν fixed. The last two estimates are explained in Section D.1.2

estimation of $\log \text{mrpk}(i)$. Following Hsieh and Klenow (2009), I use the wage bill to measure labor instead of the number of employees, as a way to account for differences in human capital. Lastly, I follow the same approach as in the baseline exercise and calibrate ν to match $\gamma = 0.1$.

Figure D.6 plots the comparison of the γ estimates from this model with the baseline estimates. For almost all countries in the sample, the estimates are smaller. Except for countries at the upper-end, this discrepancy is small. This is to be expected, as the literature has found that labor misallocation plays only a small role compared to capital misallocation in explaining overall misallocation.

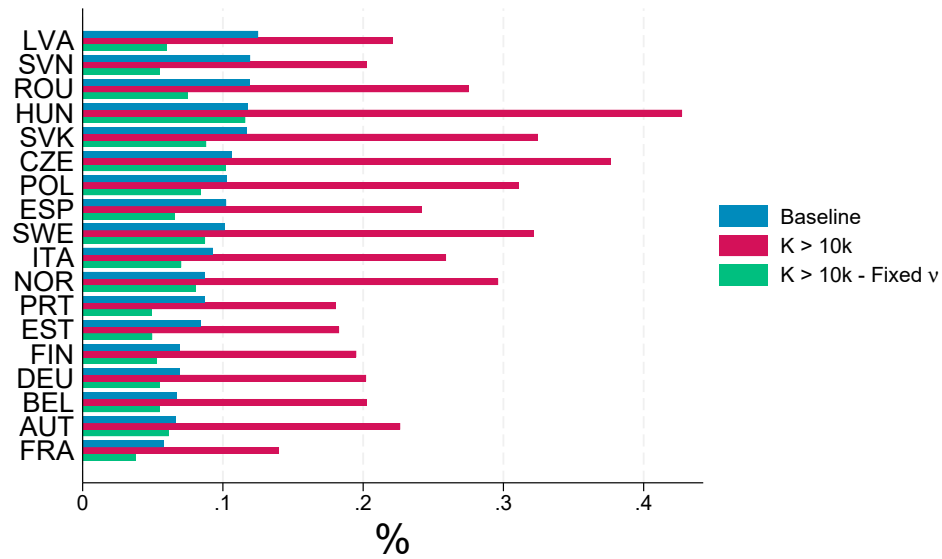
For countries in the upper end of estimates, there's a moderate reduction in the γ estimates. This does not materially affect the results in Section 6.

Figure D.3: γ Estimates for Firms with more than 5 employees



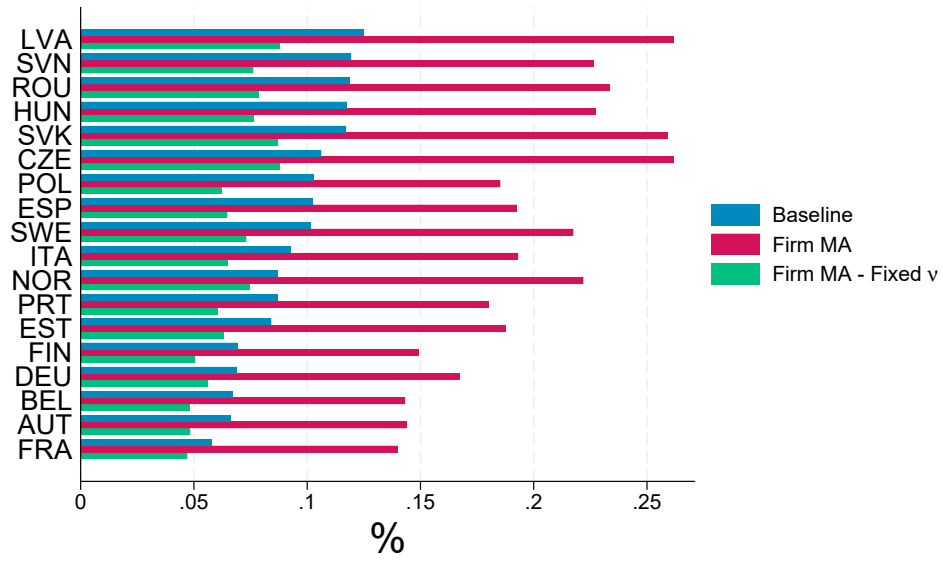
Notes: The figure plots the elasticity of TFP with respect to investment γ for a sample of countries according to (27). The blue bars plot the baseline estimates, explained in Section 6. The red bars plot the estimates using only firms with more than 5 employees to calculate capital misallocation. The green bar plots the estimates using only firms with more than 5 employees to calculate capital misallocation and keeping ν fixed. The last two estimates are explained in Section D.1.2

Figure D.4: γ Estimates Excluding Firms with Low Capital



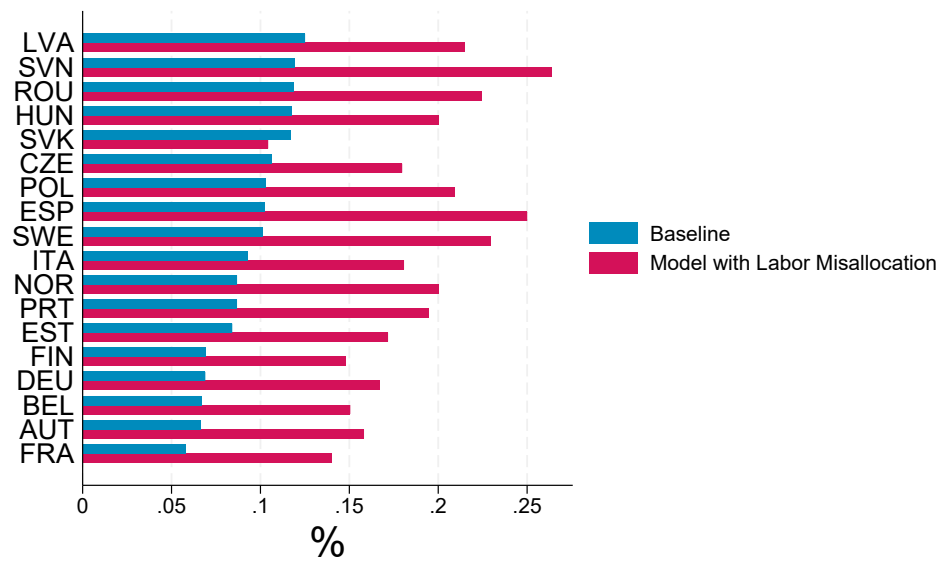
Notes: The figure plots the elasticity of TFP with respect to investment γ for a sample of countries according to (27). The blue bars plot the baseline estimates, explained in Section 6. The red bars plot the estimates using only firms which report a stock of capital larger than 10,000 euros to calculate capital misallocation. The green bar plots the estimates using only firms which report a stock of capital larger than 10,000 euros to calculate capital misallocation and keeping ν fixed. The last two estimates are explained in Section D.1.2

Figure D.5: Using Within-Firm Moving Averages to Measure σ_R^2



Notes: The figure plots the elasticity of TFP with respect to investment γ for a sample of countries according to (27). The blue bars plot the baseline estimates, explained in Section 6. The red bars plot the estimates using within-firm moving averages to calculate capital misallocation. The green bar plots the estimates using within-firm moving averages to calculate capital misallocation and keeping ν fixed. The last two estimates are explained in Section D.1.2

Figure D.6: Model with Labor Misallocation



Notes: The figure plots the elasticity of TFP with respect to investment γ for a sample of countries according to (27). The blue bars plot the baseline estimates, explained in Section 6. The red bars plot the estimates according to the model that allows for labor misallocation, derived in Section B.2.